ENGINEERING SPECIFICATION • PRODUCTION REFERENCE MANUAL

Integrated Smart Pole Subsystems & Industrial Gateway Technical Specification Manual

Comprehensive Technical Reference & Subsystem Integration Guide

PREPARED BY
ESAA Tech Team

DATE OF RELEASE
October 23, 2026

DOCUMENT VERSION
Rev 7.5 (Enterprise Edition)

DOCUMENT CLASSIFICATION
Technical Reference Standard

PHYSICAL INTERFACES
RS-485 • Gigabit LAN • 4G LTE

COMMUNICATION PROTOCOLS
Modbus RTU • A5 • SIP • TCP/IP

EXECUTIVE TECHNICAL OVERVIEW & SYSTEM SCOPE

42 Pages • 5 Modular Chapters

This document provides comprehensive, independent engineering specifications for five mission-critical subsystems deployed within municipal smart-pole infrastructure and industrial telemetry architectures. The manual details physical wiring pinouts, electrical constraints, register addresses, protocol packet structures, and direct integration guidelines with the centralized industrial gateway.

CHAPTER 1: **RS-485 3-Channel RGB LED Dimmer** — 8–26V DC input, internal bonded power bus, A5 binary protocol (Cmd 0xA0 & 0xA1), Python drivers, and strategic audit.

CHAPTER 2: **RS-485 22-Element Environmental Sensor** — Micro-meteorological transducer platform, Modbus RTU (FC 0x03), 22-register memory mapping, temperature offset formulas, and diagnostics.

CHAPTER 3: **NovaStar Taurus Multimedia Controllers** — Hardware display architecture, FPGA display engine, ViPlex/VNNOX software ecosystem, architectural gaps, and edge integration.

CHAPTER 4: **Diihoo HD IP Emergency Intercom** — Public safety alarm, 3-tier architecture, acoustic/video hardware, SIP/RTSP streaming, centralized vault recording, and API channels.

CHAPTER 5: **CJI WMD-7621U Industrial Gateway** — MediaTek MT7621A silicon, ser2net transparent bridge (Port 5021), GPIO AC relay switching, decoupled architecture analysis, and CLI checklist.

TABLE OF CONTENTS

CHAPTER 1: RS-485 3-CHANNEL CONSTANT-VOLTAGE RGB LED DIMMING CONTROLLER

1.1 Device Overview & Operational Characteristics ..... Page 4
1.2 Hardware Architecture & Internal Power Distribution (VIN+ to OUTPUT V+) ..... Page 4
1.3 Terminal Pinout & 8-Pole Wiring Assignment ..... Page 4
1.4 Electrical, Thermal & Mechanical Specifications ..... Page 5
1.5 Load Sizing & Power Budgeting Calculations ..... Page 5
1.6 RS-485 Asynchronous Serial Protocol Architecture (A5 Binary Frame) ..... Page 6
1.7 Master 3-Channel Color Synthesis (Command 0xA0) ..... Page 6
1.8 Command 0xA1 0x03: Discrete Channel Dimming ..... Page 7
1.9 Master Brightness Scaling Mathematics ..... Page 7
1.10 Autonomous Dynamic Animation Macro Engine ..... Page 7
1.11 CJI WMD-7621U Gateway Role & Technical Architecture ..... Page 8
1.11.1 MediaTek MT7621A Serial Subsystem (/dev/ttyS2) ..... Page 8
1.11.2 Software Bridging Daemon (ser2net Port 5021) ..... Page 8
1.11.3 Real-Time Packet Tracing on Port 5021 (tcpdump) ..... Page 9
1.11.4 Gateway NVRAM Configuration & Flash Persistence ..... Page 9
1.11.5 Troubleshooting Matrix & Fault Resolution ..... Page 9
1.11.6 Production Python Driver Implementation (RGBControllerCJI) ..... Page 10
1.11.7 RESTful API Architecture for SCADA & IoT Systems ..... Page 10
1.11.8 Technical Discrepancy Matrix: Modbus RTU vs. Proprietary A5 ..... Page 11
1.11.9 Strategic Procurement Mandates & Standardization Recommendations ..... Page 11

CHAPTER 2: RS-485 22-ELEMENT MULTI-PARAMETER ENVIRONMENTAL SENSOR

2.1 Multi-Parametric Transducer Profile & Specifications ..... Page 12
2.2 Transducer Specifications & Measurement Technologies ..... Page 12
2.3 Transducer Operating Principles ..... Page 13
2.4 Physical Wiring & Cable Interface ..... Page 13
2.5 Electrical Specifications & Environmental Protection ..... Page 13
2.6 RS-485 Serial Framing & Modbus RTU Protocol Standard ..... Page 14
2.7 CRC-16 Checksum Calculation Engine ..... Page 14
2.8 Complete 22-Holding Register Memory Map & Scaling Formulas ..... Page 15
2.9 Temperature Offset Formula & Conversion Callout ..... Page 16
2.10 Contiguous 21-Register Polling Transaction ..... Page 16
2.11 Microscopic Byte-by-Byte Demultiplexing Table ..... Page 16
2.12 CJI WMD-7621U Gateway Bridging Architecture ..... Page 17
2.12.1 MediaTek MT7621A Serial Subsystem Audit (3 UARTs) ..... Page 17
2.12.2 Step-by-Step Data Flow: From Physical Probe to Application ..... Page 17
2.12.3 Gateway Shell Diagnostics & Packet Capture (tcpdump) ..... Page 18
2.12.4 Gateway Configuration & Flash Persistence Architecture ..... Page 18
2.12.5 Troubleshooting Matrix & Diagnostic Remedies ..... Page 18
2.12.6 Production Python Modbus Sensor Driver ..... Page 19
2.12.7 Field Commissioning & Preventive Maintenance Checklist ..... Page 19

TABLE OF CONTENTS (CONTINUED)

CHAPTER 3: NOVASTAR TAURUS MULTIMEDIA PLAYERS & LED DISPLAY CONTROLLERS

3.1 About the Device and Product Family ..... Page 20
3.2 High-Level Hardware Architecture ..... Page 21
3.3 Factory Software Stack (As Manufactured) ..... Page 22
3.4 Hardware Specifications Matrix ..... Page 23
3.5 Third-Party Integration Capabilities ..... Page 24
3.6 Key Architectural Gaps and Vendor Dependencies ..... Page 25
3.7 Strategic Integration Approaches for Enterprise Systems ..... Page 26
3.8 Summary and Architectural Conclusions ..... Page 26

CHAPTER 4: DIIHOO HD IP EMERGENCY INTERCOM & SECURITY ALARM SUBSYSTEM

4.1 About the Device and System ..... Page 29
4.2 Complete Software Stack Overview ..... Page 30
4.3 High-Level System Architecture ..... Page 31
4.4 Hardware and Technical Specifications ..... Page 32
4.5 Centralized Recording and Evidence Management ..... Page 34
4.6 High-Level Operational Call Flow ..... Page 34
4.7 Third-Party Integration Capabilities & Ecosystem Support ..... Page 35
4.8 Security Hardening and Regulatory Standards ..... Page 38

CHAPTER 5: CJI WMD-7621U INDUSTRIAL IOT GATEWAY & INTEGRATION ENGINE

5.1 CJI WMD-7621U Industrial Gateway & Silicon SoC Specifications ..... Page 39
5.2 Gateway Operational Role & Subsystem Interconnection Topology ..... Page 39
5.3 Gateway Web Management Interface Architecture & Modules ..... Page 40
5.4 Industrial Serial Server Bridging Setup (ser2net Port 5021) ..... Page 40
5.5 Real-Time On-Gateway Hardware Verification Methods ..... Page 41
5.6 Physical Output Port & GPIO Control Architecture ..... Page 41
5.7 Hardware Analysis: Why We Can Only Physically Cut Power on Port 1 ..... Page 41
5.8 Strategic Advantages: Why a Decoupled Application Layer is Key ..... Page 42
5.9 Architectural Limitations & Field Engineering Workarounds ..... Page 42
5.10 Quick-Reference Field Administration & Diagnostic Checklist ..... Page 42

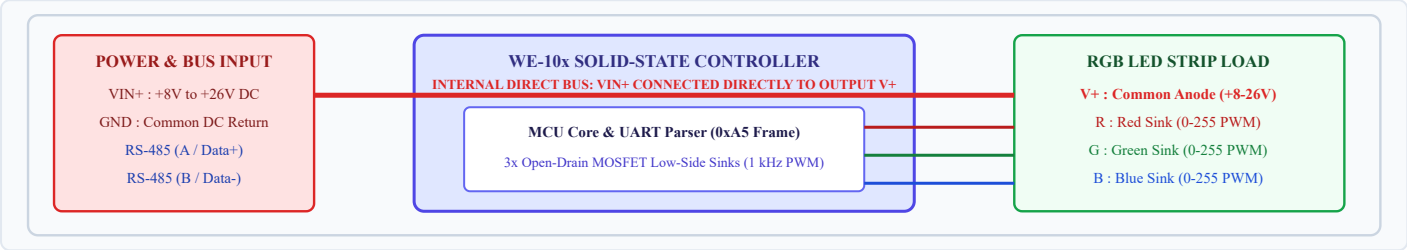
1.1 Device Overview & Operational Characteristics

The RS-485 3-Channel RGB LED Controller (WE-10x Series) is a solid-state driver engineered to regulate constant-voltage multi-color LED luminaires via High-Frequency Pulse-Width Modulation (PWM). The device integrates an RS-485 communication transceiver, an internal microprocessor, and three low-side N-channel MOSFET switching sinks.

1.2 Hardware Architecture & Internal Power Distribution

The controller is powered by a wide-range DC input from **8V to 26V DC**.

Internal Power Distribution Topology: The DC input positive terminal (`VIN+`) is **internally bonded directly to the output common anode terminal (`OUTPUT V+`)** via a heavy internal copper bus. The external RGB LED strip light receives its positive DC operating voltage directly from this internal connection. The internal microprocessor independently modulates the ground return path of each individual color channel (Red, Green, Blue) via open-drain PWM low-side sinking through three power MOSFETs.



1.3 Terminal Pinout & Wiring Assignment

Terminal Pin	Designation	Electrical Function & Wiring Definition
VIN+	DC Power In (+)	Positive DC power supply: **+8V to +26V DC** (±10%). Wire gauge: 16-18 AWG.
GND / VIN-	DC Power Return (-)	System DC Common Ground / return path for power supply.
A	RS-485 Data (+)	Non-inverting differential data line. Positive potential during Mark (idle).
B	RS-485 Data (-)	Inverting differential data line. Negative potential during Mark (idle).
OUTPUT V+	Common Anode Output	**Internally connected directly to VIN+.** Connects to common positive anode of RGB LED strip.
R	Red Channel Sink	Low-side open-drain N-MOSFET PWM sink for Red element. 0 (OFF) to 255 (Full ON).
G	Green Channel Sink	Low-side open-drain N-MOSFET PWM sink for Green element. 0 (OFF) to 255 (Full ON).
B	Blue Channel Sink	Low-side open-drain N-MOSFET PWM sink for Blue element. 0 (OFF) to 255 (Full ON).

Power Supply Voltage Matching Notice: Because `VIN+` is tied directly to `OUTPUT V+` without internal voltage conversion, the power supply voltage must match the voltage rating of the connected LED strip light (+12V DC for 12V strips, +24V DC for 24V strips).

1.4 Electrical, Thermal & Mechanical Specifications

Engineering Parameter	Design Limit / Specification	Operational Test Condition & Notes
Input Supply Voltage (VIN)	8V to 26V DC (Nominal 12V / 24V)	Absolute max: 28V DC; reverse polarity protected
Quiescent Standby Power	< 0.35 W @ 12V DC / < 0.50 W @ 24V DC	RS-485 transceiver idle, all 3 PWM outputs OFF (0%)
Max Continuous Current per Channel	4.0 A continuous per channel (R, G, B)	Low-side open-drain MOSFET; 12.0 A total maximum aggregate
PWM Modulation Frequency	1.0 kHz (±2%)	Ensures flicker-free operation and camera video compatibility
Dimming Dynamic Resolution	8-Bit Linear (256 discrete levels per channel)	16,777,216 distinct composite RGB color combinations
Operating Temperature & Ingress	-20°C to +60°C • IP20 (Indoor Rated)	Mount in vented NEMA/IP65 enclosure for outdoor luminaire duty
Galvanic Isolation	Non-isolated (Shared GND)	GND rail shared between DC input and RS-485 transceiver
Physical Dimensions & Mounting	88 mm × 38 mm × 22 mm • DIN Rail / Flange	Polycarbonate flame-retardant housing (UL94-V0)

1.5 Load Sizing & Power Budgeting Calculations

Total electrical wattage is governed by strip operating voltage (V) and linear wattage density (P_meter):

Current & Power Calculation Equations:
Total Power: $P_{total} = Length \times Linear_Wattage$ • **Total Current:** $I_{total} = P_{total} / V_{in}$ • **Per-Channel Current:** $I_{channel} = I_{total} / 3$
Voltage Drop Calculation: $V_{drop} = 2 \times I_{total} \times R_{cable} \times L_{cable}$ (Copper wire resistance: 0.013 Ω/m for 16 AWG).

Detailed Engineering Sizing Examples:

Installation Scenario	Strip Density & Length	Total Power	Supply Current (I_total)	Channel Current (I_channel)	Design Assessment
Standard +12V Strip	14.4 W/m • 5.0 meters	72.0 W	6.00 A	2.00 A	Safe: 50% of 4.0A channel limit.
High-Density +12V Strip	24.0 W/m • 4.0 meters	96.0 W	8.00 A	2.67 A	Safe: 66% of 4.0A channel limit.
Commercial +24V Strip	18.0 W/m • 10.0 meters	180.0 W	7.50 A	2.50 A	Safe: 62% of 4.0A channel limit.
Heavy Architectural +24V	28.8 W/m • 10.0 meters	288.0 W	12.00 A	4.00 A	Boundary: Exactly at 4.0A max. Recommend dual-feed.

Power Feed Best Practice: For LED runs exceeding 5 meters (12V) or 10 meters (24V), feed DC power from both ends of the strip light to prevent thermal color shifting and resistive voltage drop across the flexible PCB traces.

1.6 RS-485 Asynchronous Serial Protocol Architecture

The controller utilizes an asynchronous binary packet protocol framed with a fixed start preamble byte (`0xA5`) and dual end-of-frame terminators (`0xEE 0x5A`):

Byte Offset	Field Name	Value	Engineering Function & Framing Mechanics
Byte 0	Start Preamble	`0xA5`	Frame synchronization byte; initiates packet decoder state machine.
Byte 1	Payload Length	`0x07` or `0x0A`	Number of data bytes following until terminator bytes.
Byte 2	Target Station Address	`0xFF` / `0x01..0xFE`	0xFF = Universal Broadcast (all units); 0x01–0xFE = Unicast ID.
Byte 3	Command Opcode	`0xA0` / `0xA1`	0xA0 = Master 3-Channel Direct Update; 0xA1 = Extended / Dimming.
Byte 4..N-2	Parameters	Variable Bytes	Duty cycles (0–255), Channel selection IDs, dynamic timing parameters.
Byte N-1	Trailer Byte 1	`0xEE`	First frame delimiter byte (238 decimal).
Byte N	Trailer Byte 2	`0x5A`	Final frame terminator (90 decimal); triggers immediate PWM latch.

Dual Delimiter (0xEE 0x5A) Rationale: In industrial environments where high-current switching induces electrical transients on serial lines, requiring two consecutive specific delimiter bytes prevents noise spikes from prematurely latching false color states.

1.7 Master 3-Channel Color Synthesis (Command 0xA0)

Command `0xA0` updates all 3 color channels synchronously in a single 10-byte instruction frame:

```
FRAME STRUCTURE: [A5] [07] [ADDR] [A0] [RED: 00-FF] [GREEN: 00-FF] [BLUE: 00-FF] [00] [EE] [5A]
```

Target Color Synthesis	Raw Hexadecimal Command (10 Bytes)	Duty Cycles (R, G, B)	Visual Output Description
Pure Red	`A5 07 FF A0 FF 00 00 00 EE 5A`	(255, 0, 0)	CH1 Red 100%, CH2 Green 0%, CH3 Blue 0%
Pure Green	`A5 07 FF A0 00 FF 00 00 EE 5A`	(0, 255, 0)	CH1 Red 0%, CH2 Green 100%, CH3 Blue 0%
Pure Blue	`A5 07 FF A0 00 00 FF 00 EE 5A`	(0, 0, 255)	CH1 Red 0%, CH2 Green 0%, CH3 Blue 100%
Composite White	`A5 07 FF A0 FF FF FF 00 EE 5A`	(255, 255, 255)	All three channels 100% duty cycle (Full brightness white)
Cyan	`A5 07 FF A0 00 FF FF 00 EE 5A`	(0, 255, 255)	CH1 Red 0%, CH2 Green 100%, CH3 Blue 100%
Yellow / Amber	`A5 07 FF A0 FF FF 00 00 EE 5A`	(255, 255, 0)	CH1 Red 100%, CH2 Green 100%, CH3 Blue 0%
Magenta / Purple	`A5 07 FF A0 FF 00 FF 00 EE 5A`	(255, 0, 255)	CH1 Red 100%, CH2 Green 0%, CH3 Blue 100%
Orange	`A5 07 FF A0 FF 7F 00 00 EE 5A`	(255, 127, 0)	CH1 Red 100%, CH2 Green 50%, CH3 Blue 0%
Warm White Blend	`A5 07 FF A0 FF A0 40 00 EE 5A`	(255, 160, 64)	High-efficiency warm architectural illumination blend
All Channels OFF (Blackout)	`A5 07 FF A0 00 00 00 00 EE 5A`	(0, 0, 0)	All 3 PWM sinks OFF (Zero light emission)

1.8 Command 0xA1 0x03: Discrete Channel Dimming

To adjust an individual channel without modifying the other two channels, execute sub-command `0xA1 0x03`:

`PACKET: [A5] [07] [ADDR] [A1] [03] [CHANNEL: 1=R, 2=G, 3=B] [VALUE: 0-255] [01] [EE] [5A]`

Target Channel	Value	Raw Hexadecimal Sequence	Operational Effect
Channel 1 (Red)	100% (255)	`A5 07 FF A1 03 01 FF 01 EE 5A`	Sets Red to full 100%; leaves Green and Blue duty cycles unaffected.
Channel 1 (Red)	50% (128)	`A5 07 FF A1 03 01 80 01 EE 5A`	Dims Red to 50%; leaves Green and Blue duty cycles unaffected.
Channel 2 (Green)	100% (255)	`A5 07 FF A1 03 02 FF 01 EE 5A`	Sets Green to full 100%; leaves Red and Blue duty cycles unaffected.
Channel 2 (Green)	25% (64)	`A5 07 FF A1 03 02 40 01 EE 5A`	Dims Green to 25%; leaves Red and Blue duty cycles unaffected.
Channel 3 (Blue)	100% (255)	`A5 07 FF A1 03 03 FF 01 EE 5A`	Sets Blue to full 100%; leaves Red and Green duty cycles unaffected.
Channel 1 (Red)	0% (OFF)	`A5 07 FF A1 03 01 00 01 EE 5A`	Turns Red channel OFF; leaves Green and Blue intact.

1.9 Master Brightness Scaling Mathematics

When an operator adjusts a master brightness slider (from 0% to 100%) on a SCADA screen or Web UI, the driver calculates individual channel duty cycles using linear pulse-width scaling before packet generation:

```
scale = max(0, min(100, master_brightness_pct)) / 100.0
final_r = int(round(user_r * scale))•final_g = int(round(user_g * scale))•final_b = int(round(user_b * scale))
```

1.10 Autonomous Dynamic Animation Macro Engine

The controller incorporates internal autonomous hardware pattern generators executed directly by the internal microcontroller without continuous serial bus traffic:

Dynamic Effect Mode	Raw Hex Command Stream	Internal Execution Mechanics
7-Color Jump Flash	`A5 0A FF A1 02 06 10 10 FF 08 FF EE 5A`	Autonomous state machine sequences: Red → Green → Blue → Yellow → Cyan → Purple → White.
Breathing Light Fade	`A5 07 FF A1 01 06 00 FF EE 5A`	Internal timer executes smooth sinusoidal PWM ramping between 0% and 100% duty cycle.

> **Bus Bandwidth & Command Spacing:** While dynamic macros run autonomously, software-orchestrated color fades must enforce a minimum inter-command interval of **50 milliseconds** (maximum 20 updates/second) to prevent serial buffer overrun on the controller's 8-bit MCU.

1.11 CJI WMD-7621U Gateway Role & Technical Architecture

In industrial automation and smart building infrastructure, the RGB lighting controller connects to enterprise IP networks through the **China Jiangsu International WMD-7621U Industrial Gateway**.

Core Engineering Principle: Because high-power LED dimmers communicate via low-level differential voltage pulses over physical wires (lines A and B), they cannot communicate directly with corporate Ethernet switches, cloud servers, or SCADA software. The CJI WMD-7621U gateway acts as a high-speed, zero-latency physical-to-network bridge, translating IP packets from local Ethernet/Wi-Fi sockets directly into RS-485 serial frames.

6 Essential Technical Functions of the CJI Gateway:

- **1. Physical Media Translation:** Converts Ethernet TCP frames into RS-485 balanced serial waveforms in real time.
- **2. Hardware Direction Timing:** MediaTek silicon automatically toggles Driver Enable (DE/RE) lines without OS driver jitter.
- **3. Electrical Noise Isolation:** Isolates field wiring from enterprise IT infrastructure, absorbing inductive PWM switching noise.
- **4. Distance Extension:** Maps the serial bus to an IP socket (`192.168.11.1:5021`), extending control across campus LANs/VPNs.
- **5. Multi-System Multiplexing:** Allows multiple software clients (SCADA, scripts, mobile apps) to share lighting control.
- **6. Zero-Latency Streaming:** `ser2net` provides raw transparent stream tunneling with ~1.2 ms forwarding latency.

1.11.1 MediaTek MT7621A Serial Subsystem Architecture

The CJI WMD-7621U gateway is powered by a MediaTek MT7621A Dual-Core MIPS1004Kc processor running at 880 MHz under Linux 3.4.113 Padavan:

Linux Device Node	MMIO Base Address	Hardware IRQ	Physical Port Assignment	Operational Function & Configuration
`/dev/ttyS0`	0x1E000C00	IRQ 34	Internal Console	57600 baud (Linux boot log and diagnostic shell prompt).
`/dev/ttyS1`	0x1E000D00	IRQ 35	Serial Port 2	Managed by `ser2net_s1` (auxiliary industrial port).
`/dev/ttyS2`	0x1E000E00	IRQ 36	Serial Port 3	**Dedicated RS-485 port connected to RGB Controller. Managed by `ser2net` on TCP Port 5021.**

1.11.2 Software Bridging Daemon (ser2net) & Socket Configuration

The gateway executes the industrial `ser2net` daemon, configured to bind physical serial port `/dev/ttyS2` to network TCP port `5021`:

NVRAM Parameter	Configured Value	Engineering Purpose & Impact
`s2_enable`	`1`	Enables the serial port bridging daemon automatically on gateway boot.
`s2_dev`	`/dev/ttyS2`	Binds daemon to physical RS-485 Port 3 (MediaTek UART 2 silicon).
`s2_baud`	`9600`	Sets serial baud rate to 9600 bps to match the RGB controller standard.
`s2_databit / s2_stopbit / s2_parity`	`8 / 1 / NONE`	Standard 8N1 serial framing parameters.
`s2_port`	`5021`	Listening TCP socket port accessible across all Ethernet and Wi-Fi interfaces.

1.11.3 Real-Time Packet Tracing on Port 5021 (tcpdump)

The CJI WMD-7621U gateway includes low-level network and serial diagnostic utilities accessible via Telnet/SSH. Engineers can inspect raw lighting packets entering and exiting the gateway in real time:

```
# Execute live capture directly in the gateway shell:
tcpdump -i any -nn -X 'port 5021 and (tcp[tcpflags] & tcp-push != 0)'
```

Annotated Live Terminal Packet Trace:

```
15:10:42.102914 IP 192.168.11.2.58914 > 192.168.11.1.5021: Flags [P.], seq 1:11, ack 1
  0x0000:  4500 0033 71b2 4000 4006 e19a c0a8 0b02  E..3q.@.@.......
  0x0010:  c0a8 0b01 e622 139d 2a4f 1c90 5b82 0180  .....".*O..[...
  0x0020:  5018 01f5 6a4f 0000                      P...jO..
  0x0028:  a507 ffa0 ff00 0000 ee5a                 .........Z
--------------------------------------------------------------------------------
PAYLOAD VERIFICATION:
Offset 0x0028 reveals the exact 10 bytes: A5 07 FF A0 FF 00 00 00 EE 5A
The gateway dumped these 10 bytes directly into /dev/ttyS2 within 1.2 ms.
```

1.11.4 Gateway NVRAM Configuration & Flash Persistence

To modify or verify the serial bridging parameters in the gateway's non-volatile RAM and commit them to the SPI NOR flash partition (`/etc/storage`):

```
# Verify current settings in NVRAM:
nvram get s2_enable    # Expected output: 1
nvram get s2_port      # Expected output: 5021
nvram get s2_baud      # Expected output: 9600

# Commit settings permanently to flash:
nvram commit
/sbin/mtd_storage.sh save
```

1.11.5 Troubleshooting Matrix & Fault Resolution

Observed Symptom	Probable Root Cause	Engineering Verification & Remedial Action
No LED Response; Connection Refused	Gateway `ser2net` daemon not running or port blocked.	Telnet to `192.168.11.1`; run `netstat -tuln | grep 5021`. If absent, execute `s2_manage start`.
Packets sent OK, but LEDs remain unlit	Reversed RS-485 polarity (A and B wires swapped) or missing power.	Measure DC voltage on VIN+/GND (must be +12V/+24V). Swap lines A and B on the terminal block. Line A must be positive relative to B at idle.
Flickering or erratic color jumping	Missing common ground reference causing ground potential drift.	Ensure controller power GND is tied to gateway RS-485 shield/GND. Check that baud rate is strictly 9600 bps.
Single color channel dark (e.g. Red OFF)	Blown channel MOSFET or detached strip wire.	Measure DC voltage between Output V+ and R terminal. If stuck at VCC when Red is commanded ON, MOSFET sink is open.

1.11.6 Production Python Driver Implementation

```
import socket

class RGBControllerCJI:
    # Production driver for WE-10x 3-Channel RGB Dimmer with CJI Gateway Bridging
    def __init__(self, slave_addr: int = 0xFF, gateway_ip: str = "192.168.11.1", gateway_port: int = 5021):
        self.addr = slave_addr & 0xFF
        self.ip = gateway_ip
        self.port = gateway_port

    def build_rgb_frame(self, r: int, g: int, b: int) -> bytes:
        # Command 0xA0: Synchronous 3-Channel Color update (10 bytes)
        return bytes([0xA5, 0x07, self.addr, 0xA0, r & 0xFF, g & 0xFF, b & 0xFF, 0x00, 0xEE, 0x5A])

    def build_channel_frame(self, channel: int, value: int) -> bytes:
        # Command 0xA1 0x03: Individual Channel Dimming (1=Red, 2=Green, 3=Blue)
        ch = max(1, min(3, int(channel)))
        return bytes([0xA5, 0x07, self.addr, 0xA1, 0x03, ch, value & 0xFF, 0x01, 0xEE, 0x5A])

    def build_effect_frame(self, mode: str = "jump") -> bytes:
        # Dynamic Effects: 'jump' (7-color jump) or 'breathe' (breathing fade)
        if mode == "jump":
            return bytes([0xA5, 0x0A, self.addr, 0xA1, 0x02, 0x06, 0x10, 0x10, 0xFF, 0x08, 0xFF, 0xEE, 0x5A])
        elif mode == "breathe":
            return bytes([0xA5, 0x07, self.addr, 0xA1, 0x01, 0x06, 0x00, 0xFF, 0xEE, 0x5A])
        raise ValueError(f"Unknown effect mode: {mode}")

    def send_to_gateway(self, packet: bytes, timeout: float = 2.0) -> bool:
        # Transmits raw binary frame over TCP socket directly to CJI Gateway port 5021
        with socket.create_connection((self.ip, self.port), timeout=timeout) as sock:
            sock.sendall(packet)
        return True

# Example Dispatch: Set Color to Warm Amber Blend through China Jiangsu International Gateway
driver = RGBControllerCJI(gateway_ip="192.168.11.1", gateway_port=5021)
driver.send_to_gateway(driver.build_rgb_frame(255, 160, 64))
```

1.11.7 RESTful API Architecture for SCADA & IoT Systems

The complete integration architecture exposes HTTP REST endpoints enabling SCADA platforms, Home Assistant, and building management systems to manipulate fixtures programmatically:

Endpoint & Method	Payload Schema (JSON)	Operational Behavior & Gateway Action
`POST /api/rgb`	`{"r": 255, "g": 128, "b": 0}`	Assembles Command 0xA0 frame and dispatches over TCP port 5021 to set all 3 channels simultaneously.
`POST /api/channel`	`{"channel": 1, "value": 200}`	Assembles Command 0xA1 0x03 frame to modulate specified channel (1=R, 2=G, 3=B) without affecting others.
`POST /api/effect`	`{"effect": "jump7" | "breathe"}`	Dispatches the corresponding hardware macro sequence to initiate autonomous color cycle or breathing fade.
`POST /api/raw`	`{"hex": "A5 07 FF A0 ..."}`	Accepts arbitrary hexadecimal byte sequences for engineering diagnostic validation and low-level testing.

Production Deployment Command:
To launch the interactive Web Controller Dashboard and REST service on the supervisory server:
`python rgb_gateway_controller.py 8088` • Access GUI at `http://[SERVER_IP]:8088`.

1.11.8 Technical Discrepancy Matrix: Modbus RTU vs. Proprietary A5

During technical audits conducted by the Electric Service Administration Authority (ESAA) (documented in *RGB Controller Integration Report*), significant architectural discrepancies were documented between standard industrial Modbus RTU dimmers and the proprietary WE-10x binary protocol:

Evaluation Dimension	Standard Modbus RTU Dimmer (ZF-DIMRGB-V21)	Proprietary Protocol Driver (WE-10x Series)	Industrial Engineering Impact & Risk Assessment
Standardization	Certified Open Industrial Standard (IEC 61158 / Modbus-IDA)	Proprietary Unregistered Binary Framing (0xA5 Header)	Proprietary protocol prevents vendor interchangeability and causes lock-in.
Framing Synchronization	Standard 3.5-character silent bus idle interval ($t_{3.5} \geq 3.65$ ms)	Single start preamble byte (`0xA5`) without idle silence	Vulnerable to false synchronization if transient serial line noise mimics 0xA5.
Framing Termination	Calculated 16-Bit CRC Checksum (Polynomial 0xA001)	Fixed static end delimiters (`0xEE 0x5A`) without CRC check	Zero mathematical data validation; corrupted payload bytes are executed.
State Readback	Bi-directional telemetry via Holding Registers (0x0100–0x0108)	One-way write-only (blind dispatch; no response packet)	Master cannot verify lamp burn-out, load detachment, or thermal faults.
SCADA / PLC Native	Out-of-the-box native GUI register mapping (Siemens, Schneider, Beckhoff)	Requires custom-compiled Python wrappers and byte serial decoders	Increases software maintenance overhead and system vulnerability.
Multi-Drop Reliability	Deterministic request-response cycle prevents bus contention	Rapid broadcast flooding risks saturating bus transceivers	Requires enforced artificial delay between commands to avoid collisions.

1.11.9 Strategic Recommendations for Future Procurements

To prevent technical debt and ensure future-proof automation infrastructure, the following procurement mandates are established:

Mandate 1: Strict Contractual Submittal Compliance:
Equipment delivered to project sites must match approved engineering submittal datasheets with 100% fidelity. Delivering proprietary binary controllers when project specifications call for certified Modbus RTU hardware constitutes a critical compliance defect and introduces software fragility.

Mandate 2: Universal Standardization on Certified Modbus RTU / DMX-512 Dimmers:
All future lighting controller procurements must mandate certified Modbus RTU slaves (such as the ZF-DIMRGB-V21 series) or standard DMX-512 over RS-485. Five core engineering justifications:

1. **Universal Controller Independence:** Native plug-and-play operation with any standard industrial PLC, BMS, or SCADA platform without custom binary script wrappers.
2. **Bi-Directional Telemetry & Fault Readback:** Allows central supervisory systems to monitor actual PWM duty cycles, driver junction temperatures, and load status.
3. **Deterministic 16-Bit CRC Integrity:** Prevents false lighting transitions caused by high-power PWM inductive transients on the RS-485 differential bus.
4. **Zero-Code Commissioning:** Eliminates software development costs; field technicians commission fixtures using standard register mapping GUIs.
5. **Reduced Total Cost of Ownership (TCO):** Guarantees modular component replacement from alternative certified suppliers across the 15-year facility lifecycle.

2.1 Multi-Parametric Transducer Profile & Specifications

The RS-485 Multi-Element Environmental Sensor (HCD6815 / HCD6820 series) is an integrated micro-meteorological and air quality observation instrument. Consolidating up to 22 physical, chemical, optical, and acoustic transducers into an aerodynamically aspirated outdoor enclosure, it communicates as an autonomous Modbus RTU slave over an RS-485 serial bus.

2.2 Transducer Specifications & Measurement Technologies

Measured Parameter	Measurement Principle	Measurement Range	Accuracy & Resolution
Ambient Noise	Electret condenser acoustic microphone	30.0 to 130.0 dB	±1.5 dB • Res: 0.1 dB
Toxic Gases (H2S, SO2, NO2, CO, O3)	4-Electrode amperometric electrochemical cells	0 to 5000 ppb (gas specific)	±5% Full Scale • Res: 1 ppb
Particulates (PM1.0, PM2.5, PM10, TSP)	Laser light scattering (Mie scattering)	0 to 1000 µg/m³	±10% (±5 µg/m³) • Res: 1 µg/m³
Ambient Air Temperature	Precision PT1000 thin-film RTD	-40.00°C to +85.00°C	±0.3°C @ 25°C • Res: 0.01°C
Relative Humidity	Capacitive thin-film polymer	0.00% to 100.00% RH	±3% RH • Res: 0.01% RH
Barometric Pressure	Piezoresistive absolute silicon sensor	300.0 to 1100.0 hPa	±0.5 hPa @ 25°C • Res: 0.1 hPa
Horizontal Wind Speed	Dual-axis ultrasonic transit-time differential	0.00 to 60.00 m/s	±0.2 m/s • Res: 0.01 m/s
Horizontal Wind Direction	Ultrasonic acoustic phase differential	0.0 to 360.0 °	±3° • Res: 0.1°
Rainfall / Precipitation	Piezoelectric kinetic impact transducer	0.0 to 200.0 mm/h	±10% • Res: 0.1 mm
Solar Radiation & Illuminance	Silicon photodiode & optical pyranometer	0 to 2000 W/m² • 0 to 200,000 Lux	±5% • Res: 1 W/m² • 10 Lux
Carbon Dioxide (CO2)	Dual-beam Non-Dispersive Infrared (NDIR)	0 to 5000 ppm	±(50 ppm + 3% of reading)
Negative Oxygen Ions	Parallel-plate ion mobility capacitor	0 to 100,000 ions/cm³	±10% • Res: 10 ions/cm³
Electronic Compass Heading	3-Axis tri-axial magneto-resistive fluxgate	0.0 to 360.0 °	±2° • Res: 0.1° (Self-referencing)

Integrated Micro-Meteorological Architecture: By combining chemical cells, laser chambers, acoustic transducers, and ultrasonic flight-time sensors into a single aerodynamic chassis, external moving parts (such as spinning cups or mechanical vanes) are eliminated, preventing mechanical wear and bearing failure.

2.3 Transducer Operating Principles

Acoustic & Particulate Analysis: Ambient sound is captured by an electret microphone capsule and conditioned through an internal A-weighting frequency response network. Particulates (PM1.0, PM2.5, PM10, TSP) are measured using constant-flow laser light scattering; photodetectors sample 90-degree scattered intensity to calculate aerodynamic mass concentration per cubic meter.

Ultrasonic Wind Speed & Direction: Four ultrasonic transducers transmit acoustic pulses bidirectionally across orthogonal sound paths. Wind speed and direction are calculated from transit time differences without moving parts.

2.4 Physical Wiring & Cable Interface

Wire Color	Designation	Electrical Signal Description & Connection Target
Red Wire	VCC / Power (+)	Positive power input: **+12V to +24V DC** regulated (±10%). Max draw: 2.5W.
Black Wire	GND / Power (-)	Direct common return ground for DC power and RS-485 transceiver.
Yellow Wire	RS-485 A (D+)	Non-inverting differential data signal line. Connects to RS-485 Terminal A.
Blue Wire	RS-485 B (D-)	Inverting differential data signal line. Connects to RS-485 Terminal B.

2.5 Electrical Specifications & Environmental Protection

Electrical / Physical Parameter	Design Rating	Operational Performance Notes
Operating Supply Voltage	+12V to +24V DC (±15%)	Reverse polarity protected; over-voltage clamp diode integrated
Peak Power Consumption	< 1.5 W (Standard) / < 2.5 W (Heated)	Optional ultrasonic anti-icing heater activates below 0°C
Bus Isolation & Surge Immunity	2.5 kV RMS Galvanic Isolation	TVS surge suppressors rated to IEC 61000-4-5 (±2 kV common mode)
Ingress Protection Rating	IP65 (NEMA 4X equivalent)	UV-stabilized ASA aerodynamic housing; hydrophobic breathable vent
Operating Temperature Range	-40.0°C to +85.0°C	Operating humidity: 0% to 100% RH (fully condensing outdoor duty)
Electromagnetic Compatibility (EMC)	EN 61326-1 Industrial Immunity	Electrostatic Discharge (ESD): ±8 kV contact, ±15 kV air discharge

Installation & Mounting Notice: The sensor assembly must be mounted on a vertical pole (diameter 40–50 mm) using the supplied U-bolt brackets. The arrow indicator on the base must be aligned with True Geographic North to ensure accurate wind direction and compass heading readings.

2.6 RS-485 Serial Framing & Modbus RTU Protocol Standard

The sensor operates as a strict standard Modbus RTU Slave compliant with Modbus-IDA standards:

- **UART Framing:** 9600 Baud, 8 Data Bits, No Parity, 1 Stop Bit (**8N1**). Bit duration T_bit = 104.17 µs. Character duration T_char = 1.042 ms.
- **Inter-Frame Silence (t_3.5):** Packet boundaries are delineated by silent bus intervals ≥ 3.5 character times (t_3.5 ≥ 3.65 ms).
- **Supported Function Code: Function Code 0x03 (Read Holding Registers)**.
- **Slave Station Address:** Factory default station address is **0xFF (255 decimal)**.

2.7 CRC-16 Checksum Calculation Engine

Every Modbus RTU frame terminates with a mandatory 16-bit Cyclic Redundancy Check (CRC-16) calculated using generator polynomial `0xA001` (reversed bit representation of x^16 + x^15 + x^2 + 1).

Algorithmic Implementation Steps:
1. Initialize 16-bit register to `0xFFFF`.
2. Exclusive-OR (XOR) the lowest byte of the register with the current data byte.
3. Shift register right by 1 bit (toward LSB), zero-filling the MSB.
4. If the bit shifted out was a 1, XOR the register with polynomial `0xA001`.
5. Repeat steps 3 and 4 for all 8 bits of the current byte.
6. Repeat steps 2 through 5 for all bytes in the message frame.
7. Append the final 16-bit checksum in **Little-Endian order (Low Byte first, High Byte second)**.

```
def calculate_modbus_crc16(data: bytes) -> bytes:
    # Generator polynomial: 0xA001 (reversed representation)
    crc = 0xFFFF
    for byte in data:
        crc ^= byte
        for _ in range(8):
            crc = (crc >> 1) ^ 0xA001 if (crc & 1) else (crc >> 1)
    # Return little-endian 2-byte sequence (Low Byte, High Byte)
    return bytes([crc & 0xFF, (crc >> 8) & 0xFF])
```

Verification Test Vectors:
Query `FF 03 00 01 00 15` → Calculated CRC: `0x1280` → Appended bytes: `80 12`.
Query `FF 03 00 01 00 16` → Calculated CRC: `0x1A80` → Appended bytes: `80 1A`.

2.8 Complete 22-Holding Register Memory Map & Scaling Formulas

The internal microcontroller exposes 22 continuous 16-bit Holding Registers accessible via Modbus Function Code 0x03. All registers contain unsigned 16-bit big-endian integer values:

Reg Hex	Reg Dec	Engineering Parameter	Raw Range	Resolution / Scale	Physical Decoding Conversion Formula
`0x0001`	1	**Ambient Noise Level**	300 - 1300	0.1 dB	`Decoded = Raw / 10.0` (dB)
`0x0002`	2	**Hydrogen Sulfide (H2S)**	0 - 5000	1 ppb	`Decoded = Raw` (ppb)
`0x0003`	3	**Sulfur Dioxide (SO2)**	0 - 5000	1 ppb	`Decoded = Raw` (ppb)
`0x0004`	4	**Nitrogen Dioxide (NO2)**	0 - 5000	1 ppb	`Decoded = Raw` (ppb)
`0x0005`	5	**Carbon Monoxide (CO)**	0 - 5000	1 ppb	`Decoded = Raw` (ppb)
`0x0006`	6	**Ozone (O3)**	0 - 5000	1 ppb	`Decoded = Raw` (ppb)
`0x0007`	7	**PM2.5 Particulate**	0 - 1000	1 μg/m³	`Decoded = Raw` (μg/m³)
`0x0008`	8	**PM10 Particulate**	0 - 1000	1 μg/m³	`Decoded = Raw` (μg/m³)
`0x0009`	9	**Ambient Temperature**	0 - 12500	0.01 °C	`Decoded = (Raw / 100.0) - 40.0` (°C)
`0x000A`	10	**Relative Humidity**	0 - 10000	0.01 %	`Decoded = Raw / 100.0` (% RH)
`0x000B`	11	**Barometric Pressure**	3000 - 11000	0.1 hPa	`Decoded = Raw / 10.0` (hPa)
`0x000C`	12	**Horizontal Wind Speed**	0 - 6000	0.01 m/s	`Decoded = Raw / 100.0` (m/s)
`0x000D`	13	**Wind Direction Angle**	0 - 3600	0.1 °	`Decoded = Raw / 10.0` (Degrees °)
`0x000E`	14	**Precipitation / Rainfall**	0 - 2000	0.1 mm	`Decoded = Raw / 10.0` (mm/h)
`0x000F`	15	**Solar Radiation**	0 - 2000	1 W/m²	`Decoded = Raw` (W/m²)
`0x0010`	16	**Ambient Illuminance**	0 - 20000	10 Lux	`Decoded = Raw * 10` (Lux)
`0x0011`	17	**Ultraviolet (UV) Index**	0 - 15	1 Index	`Decoded = Raw` (UV Index Level 0-15)
`0x0012`	18	**Carbon Dioxide (CO2)**	0 - 5000	1 ppm	`Decoded = Raw` (ppm)
`0x0013`	19	**Negative Oxygen Ions**	0 - 10000	10 ions/cm³	`Decoded = Raw * 10` (ions/cm³)
`0x0014`	20	**TSP Particulate**	0 - 1000	1 μg/m³	`Decoded = Raw` (μg/m³)
`0x0015`	21	**PM1.0 Particulate**	0 - 1000	1 μg/m³	`Decoded = Raw` (μg/m³)
`0x0020`	32	**Electronic Compass Heading**	0 - 3600	0.1 °	`Decoded = Raw / 10.0` (Heading °)

Temperature Offset Formula: Register 9 uses an offset linear scale: `Decoded °C = (Raw / 100.0) - 40.0`.
Raw `6750` → `(6750 / 100.0) - 40.0 = +27.50°C` • Raw `3000` → `(3000 / 100.0) - 40.0 = -10.00°C`.

2.10 Contiguous 21-Register Polling Transaction

Industrial masters should read registers 1 through 21 (0x0001 to 0x0015) in a single contiguous block query. This eliminates multiple request round-trips and retrieves the entire environmental profile in a single 47-byte response:

1. Master Query Frame (8 Bytes):

```
Hex: FF 03 00 01 00 15 80 12
--------------------------------------------------------------------------------
Byte 0: 0xFF        -> Slave Station Address (255)
Byte 1: 0x03        -> Modbus Function 0x03 (Read Holding Registers)
Byte 2-3: 0x00 0x01 -> Start Register Address 0x0001 (Register 1: Ambient Noise)
Byte 4-5: 0x00 0x15 -> Register Quantity: 21 (0x0015) contiguous words
Byte 6-7: 0x80 0x12 -> CRC-16 Checksum (0x1280: Low byte 0x80, High byte 0x12)
```

2. Sensor Response Frame (47 Bytes Total):

```
Live: FF 03 2A 02 1A 00 00 00 00 00 00 00 00 00 00 00 0E 00 14 1A 5E
      14 50 27 D6 00 00 00 00 00 00 00 00 00 78 00 01 01 90 00 00
      00 16 00 09 B4 18
--------------------------------------------------------------------------------
Byte 0:     0xFF    -> Echo Slave Address (255)
Byte 1:     0x03    -> Echo Function Code 0x03
Byte 2:     0x2A    -> Payload Byte Count = 42 Bytes (21 registers x 2 bytes)
```

2.11 Microscopic Byte-by-Byte Demultiplexing Table

Reg	Hex Bytes	Raw Dec	Formula	Physical Decoded Result	Parameter Meaning
1	`02 1A`	538	`538 / 10.0`	**53.8 dB**	Ambient Noise Level
2-6	`00 00 ...`	0	`0`	**0 ppb**	H2S, SO2, NO2, CO, O3 (Clean air baseline)
7	`00 0E`	14	`14`	**14 µg/m³**	PM2.5 Fine Particulate
8	`00 14`	20	`20`	**20 µg/m³**	PM10 Inhalable Particulate
9	`1A 5E`	6750	`(6750/100) - 40`	**+27.50 °C**	Ambient Air Temperature
10	`14 50`	5200	`5200 / 100.0`	**52.00 % RH**	Relative Humidity
11	`27 D6`	10198	`10198 / 10.0`	**1019.8 hPa**	Barometric Pressure
16	`00 78`	120	`120 * 10`	**1,200 Lux**	Ambient Illuminance
17	`00 01`	1	`1`	**Index 1**	UV Index Level
18	`01 90`	400	`400`	**400 ppm**	Carbon Dioxide (CO2)
20	`00 16`	22	`22`	**22 µg/m³**	Total Suspended Particulate (TSP)
21	`00 09`	9	`9`	**9 µg/m³**	PM1.0 Ultrafine Particulate

Trailing CRC-16 Checksum: `B4 18` (Low: `0xB4`, High: `0x18` → Checksum Match).

2.12 CJI WMD-7621U Gateway Bridging Architecture

In automated weather monitoring stations and environmental SCADA platforms, the sensor communicates through the **China Jiangsu International WMD-7621U Industrial Gateway**:

Transparent Fieldbus Bridging Principle: The gateway operates as a transparent serial-to-network bridge rather than an application-layer display. The gateway forwards binary Modbus RTU byte streams between local Ethernet/Wi-Fi sockets and physical RS-485 balanced voltage lines with zero latency. Mathematical decoding into engineering units occurs at the supervisory application layer.

2.12.1 MediaTek MT7621A Serial Subsystem Audit

Hardware inspection of the CJI WMD-7621U running Linux 3.4.113 Padavan reveals three integrated 16550A UART blocks:

Linux Device Node	MMIO Base Address	Hardware IRQ	Physical Port Assignment	Operational Function & Configuration
`/dev/ttyS0`	0x1E000C00	IRQ 34	Internal Console	57600 baud (Linux kernel boot log and shell prompt).
`/dev/ttyS1`	0x1E000D00	IRQ 35	Serial Port 2	Managed by `ser2net_s1` (disabled by default).
`/dev/ttyS2`	0x1E000E00	IRQ 36	Serial Port 3	**Connected directly to the Environmental Sensor. Managed by `ser2net_s2` on TCP Port 5021.**

2.12.2 Step-by-Step Data Flow: From Physical Probe to Application

Step	Operational Phase	Component	Signal Mechanics & Exact System Activity
1	Transducer Sampling	Sensor Microcontroller	Probes (NTC, capacitive film, optical scattering chamber, NDIR CO2) are digitized into 16-bit registers.
2	Frame Assembly	Host Client Software	Application constructs Modbus RTU request: Slave `0xFF`, Function `0x03`, Start `0x0001`, Length `0x0015`.
3	CRC-16 Appending	Host Checksum Engine	Calculates CRC-16 (`0x1280`), appends little-endian bytes `80 12` to form an 8-byte frame.
4	TCP Ingress	Gateway Ethernet Port	Client connects via TCP to `192.168.11.1:5021` and transmits the 8-byte packet stream.
5	ser2net Forwarding	`ser2net_s2` Daemon	Daemon reads 8 bytes from socket and executes immediate `write()` to `/dev/ttyS2`.
6	RS-485 Transmission	Gateway Transceiver	UART shifts bytes onto differential lines A and B at 9600 baud (~8.3 ms duration).
7	Sensor Response	Sensor → Gateway → Host	Sensor returns 47-byte response. Gateway buffers and returns it over TCP to the client for decoding.

2.12.3 Gateway Shell Diagnostics & Packet Capture (tcpdump)

Engineers can verify environmental sensor telemetry directly inside the CJI WMD-7621U gateway shell by sniffing TCP port 5021:

```
# Live packet inspection on the CJI Gateway shell:
tcpdump -i any -nn -X 'port 5021 and (tcp[tcpflags] & tcp-push != 0)'
```

Annotated Live Acquisition Packet Trace:

```
15:15:10.420112 IP 192.168.11.2.58914 > 192.168.11.1.5021: Flags [P.], seq 1:9, ack 1
  0x0000:  4500 0030 71b2 4000 4006 e19d c0a8 0b02  E..0q.@.@.......
  0x0010:  c0a8 0b01 e622 139d 2a4f 1c90 5b82 0180  .....".*O..[...
  0x0020:  5018 01f5 6a4f 0000                      P...jO..
  0x0028:  ff03 0001 0015 8012                      ........ (Query Frame)

15:15:10.472890 IP 192.168.11.1.5021 > 192.168.11.2.58914: Flags [P.], seq 1:48, ack 9
  0x0028:  ff03 2a02 1a00 0000 0000 0000 0000 0000  ..*.............
  0x0038:  0e00 141a 5e14 5027 d600 0000 0000 0000  ....^.P'........
  0x0048:  0000 7800 0101 9000 0000 1600 09b4 18    ..x............. (47 Bytes)
```

2.12.4 Gateway Configuration & Flash Persistence Architecture

The serial bridging parameters are persisted in the gateway's NVRAM and SPI NOR flash (`/etc/storage`):

NVRAM Parameter	Configured Value	Architectural Impact on Sensor Acquisition
`s2_enable`	`1`	Starts `ser2net_s2` daemon automatically upon gateway power-up.
`s2_dev`	`/dev/ttyS2`	Binds daemon to MediaTek UART 2 (Port 3 RS-485 transceiver).
`s2_baud`	`9600`	Sets sensor communication speed (9600 bps, 8N1).
`s2_port`	`5021`	Exposes listening TCP socket port across all network interfaces.

2.12.5 Troubleshooting Matrix & Diagnostic Remedies

Observed Symptom	Probable Root Cause	Engineering Verification & Remedial Action
Connection Refused (Port 5021)	`ser2net_s2` daemon stopped or crashed.	Telnet to gateway: run `ps | grep ser2net`. If absent, execute `ser2net_s2 start`.
Request Timeout (No Response)	Reversed A/B lines or unpowered sensor.	Measure DC voltage on Red/Black wires (+12V/+24V). Verify line bias: Line A must be positive relative to Line B at idle.
CRC-16 Checksum Failure	Electrical noise or baud rate mismatch.	Verify STP cabling with shield grounded at gateway only. Confirm 120Ω end termination resistor is present.
Temperature Reads -40.0°C	Raw register 9 reads `0x0000`.	Cold boot stabilization. Sensor heating/settling cycle active. Wait 15 seconds after initial power application.

2.12.6 Production Python Modbus Sensor Driver

```
import socket, struct
from typing import Dict, Any

class EnvironmentalSensorCJI:
    # Production Modbus RTU Driver with CJI Gateway TCP Bridging
    def __init__(self, slave_addr: int = 0xFF, gateway_ip: str = "192.168.11.1", gateway_port: int = 5021):
        self.addr = slave_addr & 0xFF
        self.ip = gateway_ip
        self.port = gateway_port

    @staticmethod
    def crc16(payload: bytes) -> int:
        crc = 0xFFFF
        for b in payload:
            crc ^= b
            for _ in range(8):
                crc = (crc >> 1) ^ 0xA001 if (crc & 1) else (crc >> 1)
        return crc & 0xFFFF

    def build_poll_query(self, start_reg: int = 1, qty: int = 21) -> bytes:
        header = struct.pack(">BBHH", self.addr, 0x03, start_reg, qty)
        return header + struct.pack(" Dict[str, Any]:
        query = self.build_poll_query(start_reg=1, qty=21)
        with socket.create_connection((self.ip, self.port), timeout=timeout) as sock:
            sock.sendall(query)
            raw = sock.recv(256)
        if len(raw) < 5 or self.crc16(raw[:-2]) != struct.unpack("{raw[2] // 2}H", raw[3:3 + raw[2]])
        return {
            "noise_db": w[0] / 10.0, "h2s_ppb": w[1], "so2_ppb": w[2], "no2_ppb": w[3], "co_ppb": w[4], "o3_ppb": w[5],
            "pm2_5_ug_m3": w[6], "pm10_ug_m3": w[7], "temperature_degc": (w[8] / 100.0) - 40.0,
            "humidity_pct": w[9] / 100.0, "pressure_hpa": w[10] / 10.0, "wind_speed_m_s": w[11] / 100.0,
            "wind_direction_deg": w[12] / 10.0, "rainfall_mm": w[13] / 10.0, "radiation_w_m2": w[14],
            "illuminance_lux": w[15] * 10, "uv_index": w[16], "co2_ppm": w[17], "negative_ions": w[18] * 10,
            "tsp_ug_m3": w[19], "pm1_0_ug_m3": w[20]
        }
```

2.12.7 Field Commissioning & Preventive Maintenance Checklist

Inspection Category	Verification Procedure & Test Criterion	Remedial Action / Standard
Power Supply Quality	Measure DC voltage on Red/Black terminals under load (+12V to +24V DC). Ripple < 50 mV.	If voltage drops below 10.8V during heater cycle, increase power cable gauge (18 AWG minimum).
Bus Idle Line Bias	Measure differential voltage across A and B terminals during silent interval (t_3.5).	Differential voltage V_A - V_B must be ≥ +200 mV DC. If negative, reverse yellow and blue conductor connections.
Bus End Termination	Measure line resistance across A-B with system power completely turned OFF.	Resistance must measure between **54Ω and 62Ω** (indicating two parallel 120Ω resistors).
Physical Alignment	Inspect sensor housing orientation using bubble level and true geographic compass.	Align north orientation arrow with True Geographic North to ensure valid wind direction reporting.
Optical Particle Chamber	Quarterly visual inspection of laser scattering chamber inlet screen for insect or dust accumulation.	Gently blow dry compressed air into aspirated inlet vent. Do NOT insert cotton swabs or solvents.

TECHNICAL SPECIFICATION & HIGH-LEVEL ARCHITECTURE REPORT

NovaStar Taurus Series Multimedia Players & LED Display Controllers

Factory Product Architecture, Software Stack, Hardware Specifications, Integration Boundaries, and Architectural Gaps

Manufacturer / Developer	Xi'an NovaStar Tech Co., Ltd.
Product Family	NovaStar Taurus Series / T-Series Multimedia Controllers (TCC160, TB30, TB40, TB50, TB60, T30, T50, TU40Pro)
Subject Scope	Factory-manufactured product, native firmware, built-in software stack, and native integration boundaries

Executive Summary

This report presents a high-level architectural overview, hardware specification, native software stack review, and integration assessment of the **NovaStar Taurus Series (T-Series) LED Multimedia Players and Asynchronous Controllers**, including the TCC160, TB30, TB40, TB50, and TB60 models, manufactured by Xi'an NovaStar Tech Co., Ltd.

The Taurus family is an industrial-grade hardware platform designed to drive digital LED displays. Each controller is an integrated all-in-one solution that pairs an **Application Processor** running an embedded operating system with a high-speed **Field Programmable Gate Array (FPGA)** display engine that directly drives cascaded LED receiving cards.

Architecturally, the device as manufactured functions primarily as a **closed appliance**. Although its hardware display-scanning capability is strong, the native software ecosystem is tightly proprietary. Core operational functions, namely **content management, media authoring, dynamic scheduling, and fleet distribution**, are not exposed through open APIs. They require NovaStar's proprietary software (ViPlex Express) or a dependency on NovaStar's commercial cloud platform (**VNNOX**). Native third-party integration is limited to low-level device control commands such as brightness, power state, and diagnostics. This document outlines the product architecture and capabilities, together with the principal architectural gaps that enterprise integrators must address.

3.1 About the Device and Product Family

1.1 Commercial and Technical Identity

- **Official Manufacturer:** Xi'an NovaStar Tech Co., Ltd. (NovaStar)
- **Core Product Series:** Taurus Series / T-Series Multimedia Display Controllers

Model(s)	Description and Typical Deployment
TCC160 / TCC70	Compact core boards engineered for smart-pole screens, highway signs, and embedded kiosk displays.
TB30 / TB40 / TB50 / TB60	Boxed asynchronous multimedia players deployed across outdoor billboards, transit displays, commercial facades, and retail networks.
TU40Pro / T30 / T50	Meeting-room and conference LED display processors.

1.2 Primary Operational Roles

- **Autonomous Standalone Playback:** Runs local media loops, scheduled content, and text banners from internal solid-state flash storage without requiring an external PC at runtime.
- **Direct LED Hardware Drive:** Eliminates external video processors by sending pixel data through Gigabit Ethernet physical ports directly to downstream NovaStar receiving cards.
- **Environmental and Power Automation:** Interfaces directly with light sensors for ambient auto-dimming and with dry-contact power relays for cabinet power sequencing.

1.3 Target LED Screen Profile (320 × 280 Pixels)

In municipal, transit, and smart-pole deployments (for example emergency call stations, information bollards, and roadside kiosks), the controller frequently powers a compact, high-visibility LED canvas with the following characteristics:

- **Native Screen Resolution:** 320 × 280 pixels (89,600 physical RGB pixels in total).
- **Aspect Ratio:** Approximately 8:7, portrait-oriented, optimized for high-contrast public notifications, status indicators, and operational messaging.
- **Controller Overhead and Performance:** A 320 × 280 display consumes under 7% of the 1,300,000-pixel rated capacity of the TCC160 / TB40. This large margin supports high refresh rates, stable thermal operation, and the absence of dropped frames during media decoding.

Aspect	Summary
Autonomous Playback	Standalone local media decoding, scheduling, and rendering
Target Screen Support	Compact 320 × 280 LED canvas, and larger multi-panel setups
Hardware LED Sending Engine	Direct Gigabit PHY pixel streaming to NovaStar receiving cards
Ambient and Power Control	Optical light-sensor auto-dimming and cabinet power sequencing
Multi-Network Connectivity	Gigabit Ethernet, 2.4/5 GHz Wi-Fi, and optional 4G/5G cellular

3.2 High-Level Hardware Architecture

The Taurus architecture follows a **Dual-Engine Model**, which isolates general computing and network communication from the specialized task of driving LED display panels.

Figure 1. NovaStar Taurus controller block architecture

APPLICATION ENGINE	DISPLAY ENGINE (FPGA)
• Multi-core application processor • Hardware video decoder • Onboard RAM and flash storage • Embedded OS and system daemons	• Dedicated NovaStar display FPGA • Hardware scanning and refresh control • Integrated sending-card output ports • Colour and luminance calibration lookup

Internal video bus (Application Engine to Display Engine)

PHYSICAL INTERFACES & I/O
Network: Gigabit Ethernet (LAN/WAN), dual-band Wi-Fi, optional 4G/5G cellular **Automation:** Optical light-sensor port, programmable power relay, RS-232 / RS-485 serial **Media and Audio:** USB host ports, 3.5 mm analog audio output

Gigabit Ethernet cable

LED DISPLAY SCREEN (320 × 280)
NovaStar Receiving Card (MRV series) → 320 × 280 Physical LED Matrix

2.1 Application Engine

- **Core Responsibilities:** Handles network communication, external API requests, local file storage, and hardware video decoding (H.264/H.265).
- **Firmware Base:** An embedded operating system managed by NovaStar that runs system supervisors, watchdogs, and peripheral interfaces.

2.2 Display Engine (FPGA)

- **Core Responsibilities:** Acts as an integrated hardware sending card. It receives decoded video frames from the application processor, maps them to the physical cabinet coordinates, and transmits high-speed serial pixel streams over Gigabit Ethernet directly to the receiving cards.
- **Receiving Card Management:** Connects to downstream receiving cards (for example the MRV series) mounted inside the LED cabinet, which in turn drive the physical 320 × 280 LED matrix.

3.3 Factory Software Stack (As Manufactured)

NovaStar delivers the Taurus controller with a pre-installed software suite. The stack separates basic device control from content management.

#	Component	Role	Description
1	Embedded OS and Firmware	Device Platform	Hardened embedded OS with NovaStar drivers for display hardware, watchdogs, and network interfaces.
2	Taurus Core Services Daemon	Device Control Plane	Background services managing device lifecycle, hardware states, and display engine synchronization.
3	Built-in Local REST Service	Device Control API	Embedded HTTP/HTTPS service (Port 16674) providing basic device-level control (brightness, power, reboot, telemetry).
4	NovaPlayer / ViPlex Player	Factory Playback Engine	Built-in proprietary media player that plays NovaStar-formatted playlists and schedules.

#	Component	Role	Description
5	ViPlex Express (PC application)	Local Display Manager	Proprietary Windows desktop suite for content creation, screen layout partitioning, and local network publishing.
6	NovaLCT Engineering Suite	Hardware Calibration Tool	Engineering software for cabinet mapping, receiving-card configuration, and seam/chroma calibration.
7	VNNOX Cloud Platform	Proprietary Cloud CMS	NovaStar's multi-tenant commercial cloud service for remote content scheduling, advertising campaigns, and fleet updates.

3.1 Content Management and Playback

- **Proprietary Player (NovaPlayer):** The built-in playback application is designed exclusively to execute playlists authored through NovaStar's own software tools. It processes custom visual programs with predefined windows, text overlays, and media sequences.
- **ViPlex Express:** The official desktop software for local, offline content authoring and publishing. Operators use it to divide the screen into zones, create schedules, and publish them to controllers across the local network.

3.2 Engineering and Calibration

- **NovaLCT:** The primary technical tool used during installation and commissioning. It is not used for daily media management, but is essential for configuring display dimensions (for example 320 × 280), loading cabinet configuration files, and performing initial display calibration.

3.3 Remote Cloud Management

- **VNNOX Cloud Platform:** NovaStar's proprietary digital signage SaaS. Advanced capabilities, including scheduled advertising distribution across multiple remote screens, wide-area remote updates, proof-of-play logs, and remote status monitoring over 4G/WAN, are delivered through VNNOX.

3.4 Hardware Specifications Matrix

Subsystem	Specification Details
Manufacturer	Xi'an NovaStar Tech Co., Ltd.
Product Series	Taurus T-Series (TCC160 / TB30 / TB40 / TB50 / TB60)
Processor Architecture	Industrial multi-core application processor
Video Decoding	Hardware decoding for H.264, H.265 (HEVC), VP9, up to 4K
Onboard Storage	8 GB - 32 GB eMMC solid-state storage
Display Engine	Proprietary NovaStar FPGA with integrated sending card
Target Screen Resolution	**320 × 280 pixels** (89,600 pixels total; native portrait canvas)
Max Controller Capacity	TCC160 / TB40: up to 1,300,000 pixels TB30: up to 650,000 pixels TB60: up to 2,300,000 pixels
Physical Display Ports	Gigabit Ethernet RJ-45 output (to downstream receiving cards)
Network Interfaces	10/100/1000 Mbps RJ-45 Ethernet port (LAN/WAN)
Wireless Connectivity	Dual-band Wi-Fi (2.4 GHz / 5 GHz); concurrent AP and Client modes
Cellular Expansion	Optional 4G / 5G modem slot (with onboard SIM tray)

Subsystem	Specification Details
Industrial Bus	1 x RS-232 / RS-485 serial interface for external sensors or PLC control
Sensor Port	1 x dedicated optical light-sensor interface (automatic dimming)
Power Relay	1 x integrated dry-contact power relay for cabinet power sequencing
USB Connectivity	USB host ports for direct USB media playback and configuration
Audio Output	1 x 3.5 mm analog stereo audio output
Operating Temperature	Industrial rating: -20 °C to +60 °C (-4 °F to +140 °F)
Compliance and Standards	CE, FCC, RoHS, CCC, UL/CB, ISO 9001:2015

3.5 Third-Party Integration Capabilities

A common point of confusion when evaluating NovaStar screens is the scope of native integration available out of the box. As manufactured, **native integration is strictly limited to device-level hardware controls**. There is no open API for content scheduling or playlist management.

Figure 2. Native integration scope (as manufactured)

Available Native APIs (Device Hardware Control Only)	Unavailable via Open APIs (Proprietary / Closed)
• Screen power on / off (blackout / normal)	• Dynamic content scheduling APIs
• Brightness adjustment (percent / raw 0-255)	• Open playlist creation / editing
• Device reboot / restart	• Media queue and campaign management
• Volume level control	• Proof-of-play audit APIs
• Receiving-card telemetry (temperature / voltage)	• Multi-screen remote sync protocols
• Raw file upload and app install endpoint	• Direct framebuffer ingestion

5.1 What the Native System Provides

Third-party platforms communicating with the device over the local network, via the embedded REST service on Port 16674, can access the following:

1. **Screen State Control:** Remotely set the screen to display normally, black out, or freeze.
2. **Brightness Management:** Programmatically read and set screen brightness, either as a percentage or as a raw 0-255 value.
3. **Hardware Lifecycle:** Trigger system reboots and query device uptime and firmware version.
4. **Receiving Card Telemetry:** Query downstream receiving-card health, including operating temperatures, voltages, and connection link status.
5. **Serial Passthrough:** Transmit and receive commands via the RS-232/RS-485 port for external sensor integration.
6. **Package Upload and Silent App Installation:** The device provides an endpoint (`/terminal/emergency/v1/app/install`) that allows administrative deployment of software packages directly to the controller.

5.2 What the Native System Does Not Provide

- There is **no open REST API to schedule content**, define playlists, or trigger specific media playback dynamically.
- Third-party software cannot instruct the native player to display an arbitrary image, video, or web feed through simple standard API parameters.
- Content management remains locked inside NovaStar's proprietary software workflows.

3.6 Key Architectural Gaps and Vendor Dependencies

For organizations seeking to embed NovaStar screens into enterprise command centers, smart-city dashboards, or third-party digital signage platforms, the following architectural gaps represent the primary integration obstacles.

Gap	Summary
Gap 1: Proprietary Content Stack	Playlist creation, multi-zone layout, and scheduling are completely closed within NovaStar proprietary tools.
Gap 2: Heavy Cloud Dependency (VNNOX)	Centralized scheduling, remote distribution, fleet monitoring, and multi-display sync depend on NovaStar's commercial cloud SaaS platform.
Gap 3: Minimal Native Integration Surface	The native API is strictly a hardware control interface (power, brightness, reboot). Beyond basic device parameters there are no open programmatic hooks for content.

Gap 1: Content Management and Scheduling Are Completely Proprietary

- **The Constraint:** In digital display deployments, the two most critical operational tasks are content management and scheduling. On the Taurus platform as manufactured, both are entirely proprietary.
- **The Reality:** The built-in player does not accept standard playlist formats or open scheduling instructions. Content authoring is tethered to NovaStar's proprietary software (ViPlex Express or VNNOX). A third-party platform cannot natively push a playlist or command the screen to play specific media through standard REST calls.

Gap 2: Heavy Vendor Cloud Service Dependency (VNNOX Lock-in)

- **The Constraint:** Enterprise features required to manage a distributed screen network are architected around NovaStar's commercial cloud SaaS, VNNOX.
- **The Reality:** Capabilities such as remote advertising distribution across geographic regions, multi-screen synchronized playback, proof-of-play auditing, and centralized over-the-air firmware upgrades depend on connecting the devices to NovaStar's cloud infrastructure.
- **The Impact:** Organizations operating within sovereign networks, high-security facilities, or unified municipal operations centers cannot easily host an on-premises equivalent of VNNOX. They must either rely on a third-party vendor cloud or forgo the screen's built-in remote management features.

Gap 3: Highly Restricted Native Integration Scope

- **The Constraint:** The native integration surface exposed by the controller is limited to basic device hardware commands.
- **The Reality:** While the REST API functions reliably for switching the display on/off, adjusting brightness, and reading temperature metrics, it provides **no native hooks into the content pipeline**. Third-party platforms cannot achieve true end-to-end screen automation (for example dynamically displaying an emergency alert banner or a live operational KPI feed) through the factory API alone.

3.7 Strategic Integration Approaches for Enterprise Systems

Because the hardware display engine is capable while the content software stack is closed and cloud-dependent, enterprise integrators typically adopt one of two architectural strategies.

Strategy	How It Works	Trade-Offs and Best Use Case
Approach A: Hardware Control Only	A third-party system uses the local REST API strictly for power, brightness, and health telemetry. Content remains in ViPlex/VNNOX.	Simple and officially supported. Content and schedules must still be managed manually via ViPlex or VNNOX. Suitable when a unified CMS is not required.
Approach B: Platform Independence (Edge Player App)	A third-party deploys a dedicated, lightweight edge player application onto the device using the native app install API, replacing NovaPlayer.	Eliminates VNNOX and ViPlex entirely. Full control over content and schedules. The hardware display engine (FPGA) is fully preserved for LED scanning.

7.1 Approach A: Hardware Control Only (Accepting the Vendor Ecosystem)

- Suitable for projects in which operators are comfortable using NovaStar's ViPlex Express or VNNOX for all media creation and scheduling, but require an overarching dashboard (for example a smart-city control room) to monitor screen health, automate brightness dimming, and control power states.
- Implementation is straightforward using the built-in REST API on Port 16674.

7.2 Approach B: Full Platform Independence via Native Edge App Deployment

This approach is suitable for organizations that require a **single, unified enterprise platform** for both screen control and content scheduling, without reliance on NovaStar's proprietary software or cloud services. It operates as follows:

1. The integrator uses the native `/terminal/emergency/v1/app/install` endpoint to deploy a custom, lightweight playback application directly onto the controller.
2. The custom application is configured to launch on boot, taking over the display output.
3. The custom player communicates directly with the enterprise's private backend (for example via private object storage or local APIs) for media assets and schedules.
4. The underlying NovaStar hardware and FPGA continue to handle physical LED module driving, giving the organization the full benefit of NovaStar's display engineering while bypassing vendor software and cloud lock-in.

3.8 Summary and Architectural Conclusions

When evaluated **as manufactured by NovaStar**, the platform is assessed as follows:

Area	Assessment	Rationale
Hardware and Physical Display Driving	**Good**	The dual-engine architecture with a dedicated FPGA provides stable, high-capacity LED control, seamless receiving-card management, and reliable peripheral support.
Native Integration Surface	**Extremely Limited**	Confined strictly to basic hardware-level device commands (brightness, power, reboot, telemetry).
Content Management and Scheduling	**Proprietary and Cloud-Dependent**	Core functions are locked within NovaStar's proprietary desktop application (ViPlex Express) and cloud infrastructure (VNNOX).

For enterprise solutions, recognizing this boundary is critical. While the device excels as an LED display engine, integrating it into an open, third-party operational ecosystem requires either accepting a split workflow (using vendor tools for content) or deploying a custom edge player to achieve complete platform independence.

TECHNICAL SPECIFICATIONS

Product Overview, System Architecture and Technical Specifications

Diihoo HD IP Emergency Intercom & Security Alarm System

Product Overview, High-Level Architecture, and Technical Specifications

Manufacturer / Developer	Zhejiang Diihoo Technology Co., Ltd.
System Designation	Diihoo HD IP Intercom Alarm System (Version 2.x Series)
Date of Release	October 2026

Executive Summary

This report presents a high-level operational overview, system architecture, and comprehensive hardware and software specifications for the **Diihoo HD IP Emergency Intercom & Security Alarm System**, manufactured by **Zhejiang Diihoo Technology Co., Ltd.**

The Diihoo HD IP Intercom System is an industrial-grade, network-native emergency communication and situational awareness platform engineered for critical public safety, transit, municipal, and commercial installations. The system provides instantaneous two-way voice communication, high-definition real-time video surveillance, public address voice paging, remote access control actuation, and centralized automated audio and video recording.

Built upon a flexible multi-tier architecture, the platform encompasses outdoor vandal-resistant intercom terminals, a centralized alarm management server, dedicated dispatcher workstations, electronic mapping, large-scale TV video wall controllers, and modern web integration gateways. This document outlines the system's core operational features, complete software suite, high-level architecture, device specifications, and regulatory standards compliance.

Contents

Section	Title
4.1	About the Device and System
4.2	Complete Software Stack Overview
4.3	High-Level System Architecture
4.4	Hardware and Technical Specifications
4.5	Centralized Recording and Evidence Management
4.6	High-Level Operational Call Flow
4.7	Third-Party Integration, Ecosystem Support and Operational Considerations
4.8	Security Hardening and Regulatory Standards

4.1 About the Device and System

1.1 Commercial and Technical Identity

Item	Detail
Official Brand Name	Diihoo
Manufacturer	Zhejiang Diihoo Technology Co., Ltd.
Product Line	Diihoo High-Definition IP Emergency Intercom Terminal Series
Software Ecosystem	Diihoo HD IP Intercom Alarm System (Enterprise Edition)
Core Operating Platform	Hardened Industrial Embedded Linux

1.2 Operational Purpose and Applications

The Diihoo IP Emergency Intercom Terminal series is engineered to provide reliable, instant communication between individuals in the field and central security personnel, particularly during emergency or assistance situations.

Primary Deployment Scenarios

- **Smart Cities and Municipal Public Safety:** street-level emergency call pillars, pedestrian malls, public parks, and underpasses.
- **Mass Transit and Transportation Hubs:** train platforms, bus interchanges, airport terminals, and multi-story parking facilities.
- **Commercial, Educational and Healthcare Campuses:** school walkways, hospital emergency entrances, perimeter checkpoints, and parking structures.
- **Critical Infrastructure and Industrial Sites:** highway emergency call boxes, energy substations, automated vehicle entry gates, and manufacturing complexes.

1.3 Key Operational Capabilities

- **Configurable Multi-Button Call Assistance:** terminals support customizable physical push buttons (single-button or multi-button variants) mapped to distinct functional roles, namely an Emergency / SOS button (instant high-priority dispatch to the security control center with automatic alarm chime and camera pop-up), an Information / Assistance button (general inquiry, reception, or customer assistance speed-dial), and a Staff / Security direct-dial button (dedicated lines to guardhouses or building managers).
- **High-Definition Real-Time Video:** an integrated wide-angle Full HD camera provides instant visual confirmation to operators upon call initiation, enabling rapid situational assessment.
- **Full-Duplex Hands-Free Voice:** high-fidelity two-way audio featuring hardware-accelerated Acoustic Echo Cancellation (AEC) and Active Noise Suppression (ANS), delivering clear speech intelligibility in loud environments.
- **Public Address (PA) and Area Broadcast:** supports live dispatcher announcements and emergency civil alarm broadcasts to individual terminals or pre-configured terminal zones simultaneously.
- **Remote Peripheral and Access Control:** built-in digital output relays allow dispatchers to remotely operate automated vehicle barrier gates, electromagnetic door strikes, and warning strobe lights directly from the monitoring console.
- **Centralized Server Recording:** every call, alarm event, and two-way voice session is automatically captured and preserved on the central management server for evidence review and audit compliance.

4.2 Complete Software Stack Overview

The Diihoo IP Intercom ecosystem is delivered as a comprehensive, enterprise-ready software suite designed to cover all operational aspects, from device provisioning to command center dispatching.

No.	Module	Function
1	Diihoo AlarmServer (Core PBX & Management Server)	Central PBX engine, device registration, alarm processing, and automated server-side recording service.
2	Diihoo AlarmClient (Desktop Dispatch Console)	Native security dispatcher workstation for call handling, multi-channel video viewing, push-to-talk, and barrier control.
3	Diihoo AlarmTvWall (Video Wall Management)	Multi-screen TV wall management suite for command and control centers, supporting matrix switching and multi-stream grids.
4	Diihoo AlarmSetup (Device Configuration Tool)	Hardware provisioning, network subnet discovery, audio gain calibration, and field maintenance utility.

No.	Module	Function
5	Diihoo AlarmServerManager & Access2Mysql	System administrator utility for network adapter bindings, port range assignments, and enterprise database migration.
6	Diihoo Web Dispatcher & Gateway (Web & Integration Layer)	Cross-platform browser console and WebSocket/REST API gateway for third-party VMS / PSIM integration.
7	Electronic Map (E-Map / GIS) Subsystem	Interactive campus and floor plan navigation with real-time device status badges and emergency call location pop-ups.

2.1 Module Descriptions

Diihoo AlarmServer. The central operational core of the platform. It maintains real-time keepalive heartbeats with all deployed terminals, enforces access permissions, coordinates call routing, logs system events, and automatically captures high-definition video and audio recordings to the server's local or networked disk storage.

Diihoo AlarmClient. The primary graphical desktop workstation used by security operators. It features one-click call pickup, multi-window video monitoring, two-way push-to-talk (PTT) communication, peripheral trigger buttons, and historical log auditing.

Diihoo AlarmTvWall. Designed for large-scale security operations centers (SOC). It decodes multiple intercom camera streams simultaneously across professional display walls and HDMI matrix decoders.

Diihoo AlarmSetup. A field engineering utility that broadcasts network discovery frames to locate, configure, and calibrate terminal IP addresses, server targets, microphone gains, and loudspeaker volume levels without requiring physical dismounting.

Diihoo AlarmServerManager & Access2Mysql. Administrative management tools that allow administrators to configure server network cards and communication ports, and to scale from local database storage to enterprise database clusters (such as MySQL).

Diihoo Web Dispatcher & Integration Gateway. A zero-install, responsive web client providing full dispatching capabilities across modern desktop and mobile browsers, coupled with an integration gateway for third-party Video Management Systems (VMS).

Electronic Map (E-Map) System. A multi-level interactive GIS/CAD mapping module displaying terminal placements across facility blueprints. When an emergency call is triggered, the map automatically centers on the active terminal and highlights its physical location.

4.3 High-Level System Architecture

The Diihoo platform utilizes a modular, multi-tier architecture ensuring robust separation between field hardware, server management, and dispatcher presentation layers.

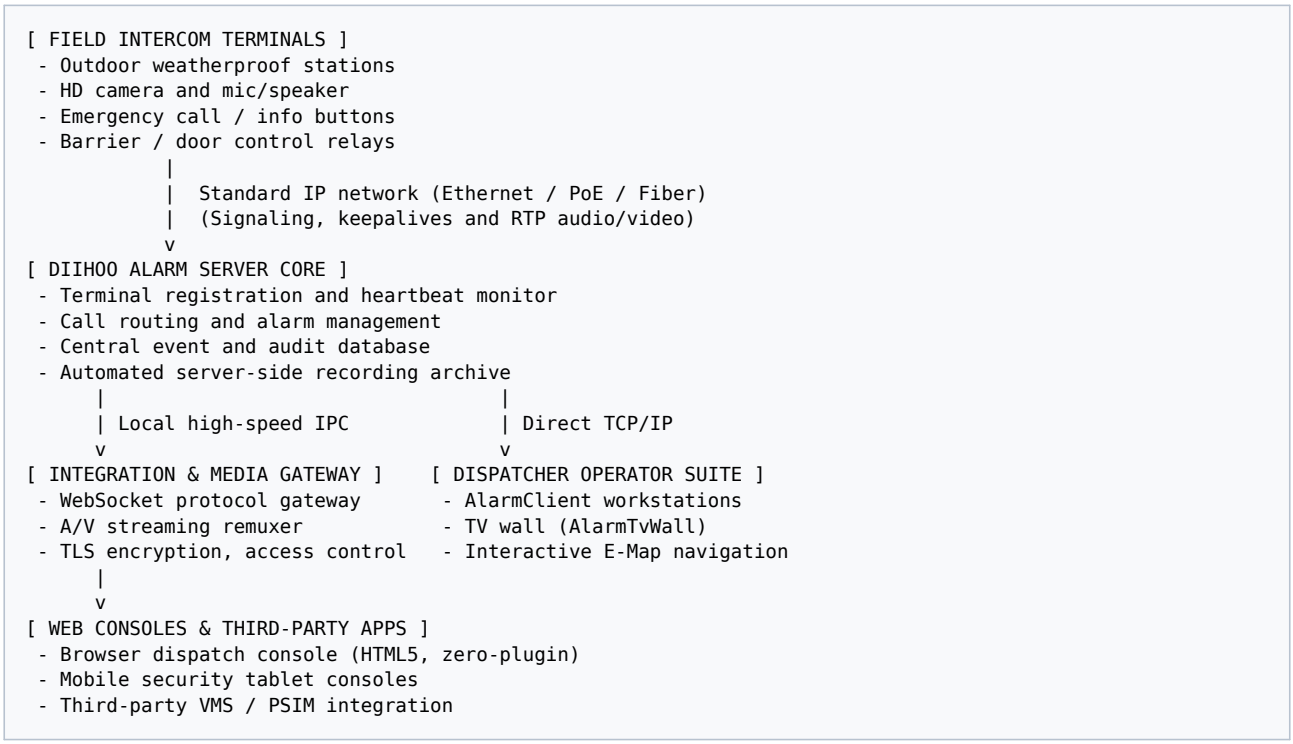


Figure 1. High-level system architecture

3.1 Tier Breakdown

Edge Terminal Tier (Field Layer). Hardened intercom stations are deployed in outdoor or indoor zones. They interface with physical push buttons, video optics, acoustic components, and dry-contact barrier relays, and communicate continuously with the central server over IP using standard Ethernet network cabling.

Server Management Tier (Central Core). This tier hosts the central Diihoo PBX engine (AlarmServer). It manages terminal state tables, priority call routing, user authorization, and system event journals. When a call or emergency button is activated, the server automatically captures the live audio and video stream and saves the record to the server's storage repository. An automated disk management routine periodically purges the oldest archived media to ensure uninterrupted recording continuity.

Gateway and Integration Tier (Connectivity Layer). This tier bridges traditional telephony and video streams into standardized, web-friendly WebSocket and REST protocols. It enables secure remote dispatching from modern web browsers without proprietary plugins, and exposes integration endpoints for connecting the intercom network into enterprise Command and Control (C2), VMS, or Physical Security Information Management (PSIM) software.

Operator Presentation Tier (Dispatch Layer). This tier provides dispatch personnel with multiple operational interfaces: native desktop workstations (AlarmClient), wall-mounted operations displays (AlarmTvWall), and flexible browser-based consoles.

4.4 Hardware and Technical Specifications

The following table provides the technical specifications for the Diihoo IP Emergency Intercom Terminal series.

Category	Parameter	Specification
System Platform	Processor Architecture	High-performance 32-bit embedded RISC processor
	Operating System	Hardened embedded Linux OS
	Watchdog	Dual hardware and software watchdog for automatic recovery
Video & Optics	Optical Sensor	1/2.8" progressive scan CMOS sensor
	Lens & Field of View	2.8 mm wide-angle lens (110° horizontal field of view)
	Video Resolution	Full HD 1080p (1920 × 1080) / HD 720p dual-stream
	Video Compression	H.264 / AVC (Baseline and Main Profiles)
	Frame Rate	25 fps / 30 fps (real-time smooth video)
	Low-Light Sensitivity	0.01 Lux (color); 0.001 Lux (black & white mode)
	Day / Night Control	Automatic IR-cut filter removal (ICR) with digital WDR
Audio Subsystem	Audio Input	Integrated high-sensitivity omnidirectional microphone
	Audio Output	Moisture-sealed, weatherized high-output loudspeaker
	Sound Output Power	Standard intercom: 3 W RMS; public address boost: up to 10 W
	Sound Pressure Level	≥ 85 dBA at 1 meter (audible in noisy outdoor areas)
	Audio Sampling Rate	8 kHz / 16 kHz linear PCM
	Acoustic Processing	Hardware AEC, noise suppression (ANS), AGC
Physical Controls & I/O	Call Push Buttons	Single or multi-button configurations (SOS, Info, Speed-Dial)
	Button Durability	Vandal-resistant stainless steel push buttons (≥ 1,000,000 operations)
	Digital Output Relays	2 × Form C dry-contact relays (door strike, barrier gate, strobe siren)
	Digital Sensor Inputs	2 × optocoupler-isolated alarm inputs (tamper switch, door reed contacts)
	Visual Status Ring	Multi-color LED status ring (standby, ringing, call connected)
Networking	Physical Port	1 × RJ-45 10/100 Mbps auto-negotiating Fast Ethernet
	Protocols Supported	TCP/IP, UDP, RTP, HTTP, RTSP, SIP 2.0 compatibility, DHCP, NTP
	Server Compatibility	Fully managed via Diihoo AlarmServer suite
Power & Electrical	Operating Voltage	12 V DC (±10%) standard input
	Power over Ethernet	IEEE 802.3af Class 3 PoE
	Power Consumption	Idle: ≤ 3.5 W; active call and video: ≤ 7 W; PA peak: ≤ 12 W
	Surge & Lightning	Built-in 6 kV lightning, surge, and ESD protection
Mechanical & Environmental	Housing Enclosure	Brushed 304/316 stainless steel faceplate with cast alloy backbox
	Ingress Protection	IP65 / IP66 certified (waterproof and dustproof)
	Vandal Resistance	IK08 / IK10 certified (high mechanical impact resistance)
	Operating Temperature	-30 °C to +65 °C (-22 °F to +149 °F)
	Operating Humidity	10% to 95% relative humidity (non-condensing)

Category	Parameter	Specification
	Installation Method	Flush-mount (wall/bollard) or surface-mount with weather hood

4.5 Centralized Recording and Evidence Management

A critical requirement for public safety and regulatory compliance is the integrity, reliability, and accessibility of call records and evidentiary data.

- **Automatic Call Session Capture:** the system is engineered to capture audio and video centrally. When an emergency call or alarm is initiated from any field terminal, Diihoo AlarmServer automatically records the live camera video stream and the full-duplex two-way voice conversation directly onto the host server's local or networked disk storage.
- **Storage Architecture and the “SD Playback” Misnomer:** in official vendor dialogs and legacy documentation, recording menus are labeled “SD Playback”. In field operations, this frequently causes the misconception that recordings reside on physical MicroSD cards inside the terminal. In reality, the terminal streams live RTP media over the network, and all recordings are written directly to the central server storage repository (e.g., `C:\ALMSERVER\RECORD\`). Terminals in this deployment do not require or utilize edge flash storage.
- **Proprietary Recording Container Format (.bin):** the native server-side recorder outputs proprietary container files (`AS_<DEVICE_ID>_<INDEX>.bin`) rather than open multimedia formats such as MP4 or AVI.

Operational Implication. These archives cannot be opened directly with generic commercial media players (e.g., VLC or Windows Media Player). They must be audited and reviewed through the official Diihoo desktop suite (AlarmClient) or decoded using the vendor's DirectShow filters (`DDMediaFilter.dll`).

Web Client Call Recording. Because the native server software does not export standard web video formats out of the box, modern web dispatcher interfaces incorporate client-side HTML5 recording capabilities (MediaRecorder API), allowing operators to export standard, universally playable .mp4 or .webm call clips during an active session.

- **Intelligent Capacity Management (Circular FIFO):** the server features an automated disk storage monitoring routine. The system continuously evaluates all available logical drives and directs recordings to the volume with the largest available free space (requiring a minimum threshold greater than 1.0 GB). A background maintenance cycle ensures that if available capacity falls below a configured threshold (typically 10 GB), the oldest archived recordings are automatically purged in circular FIFO order, preventing server disk exhaustion.
- **Audit Trail and Call Detail Logging:** in addition to media recordings, the system logs every system action, including call setup times, call durations, operator responses, and peripheral barrier activations, into a relational database for historical analysis and regulatory reporting.

4.6 High-Level Operational Call Flow

The standard operational sequence during an emergency event follows a streamlined five-stage workflow.

Stage	Phase	Description
1	Initiation	A citizen or user presses the emergency call button on the outdoor terminal. The terminal visual indicator illuminates and plays a local ringback chime, and a high-priority alarm packet is transmitted across the IP network to the server.
2	Server Routing and Broadcast	Diihoo AlarmServer processes the incoming call event, indexes the call in the event database, engages the server-side recording engine to capture live video and audio, and dispatches the call alert simultaneously to operator workstations and video walls.
3	Dispatcher Response	The security operator acknowledges and answers the call. Live 1080p video from the terminal camera appears instantly on the dispatcher's screen, two-way full-duplex audio is opened, and, if configured, the electronic map centers on the terminal's location.
4	Peripheral Actuation (Optional)	If required (for example, verifying an emergency vehicle or authorized visitor), the dispatcher clicks a control button on the workstation screen and the terminal actuates its internal relay to raise a vehicle barrier or unlock a door.
5	Call Termination and Archive	The operator concludes the call. The terminal status returns to normal standby, and the server finalizes and indexes the recorded audio/video file for future retrieval.

4.7 Third-Party Integration Capabilities & Ecosystem Support and Operational Considerations

An essential consideration for municipal infrastructure, smart city platforms, and security command centers is whether the Diihoo IP Intercom system can be integrated into third-party software (such as Video Management Systems, Physical Security Information Management platforms, mobile apps, or custom web dashboards), and what official support exists.

7.1 Integration Feasibility and Support Status

Integration Capability: capable across multiple tiers. The hardware endpoints and backend servers support standard IP streaming and signaling, while gateway components allow modern web and software integration.

Official Vendor Support: Zhejiang Diihoo Technology Co., Ltd. provides official integration components, including the following.

- **Diihoo Web SDK (`dhsdk / dhsdk.min.js`):** client integration library facilitating session registration, call answering, two-way audio streaming, and peripheral controls.
- **Native Platform SDK (`DDPlatform.dll` and C++ libraries):** Windows native communication libraries for building custom C++, C#, or .NET applications interacting with AlarmServer.
- **WebSocket Integration Gateway (`intercomMain`):** dedicated integration daemon converting internal PBX events and binary media streams into standard JSON-RPC text frames and binary WebSocket payloads.
- **Open Protocol Compliance:** the hardware supports standard RTSP (Real-Time Streaming Protocol) and SIP 2.0 (Session Initiation Protocol), allowing third-party platforms to interface with terminals directly without vendor lock-in.

7.2 Integration Approaches and Architecture

Third-party platforms can integrate with the Diihoo ecosystem using three primary approaches, depending on the technical requirements.

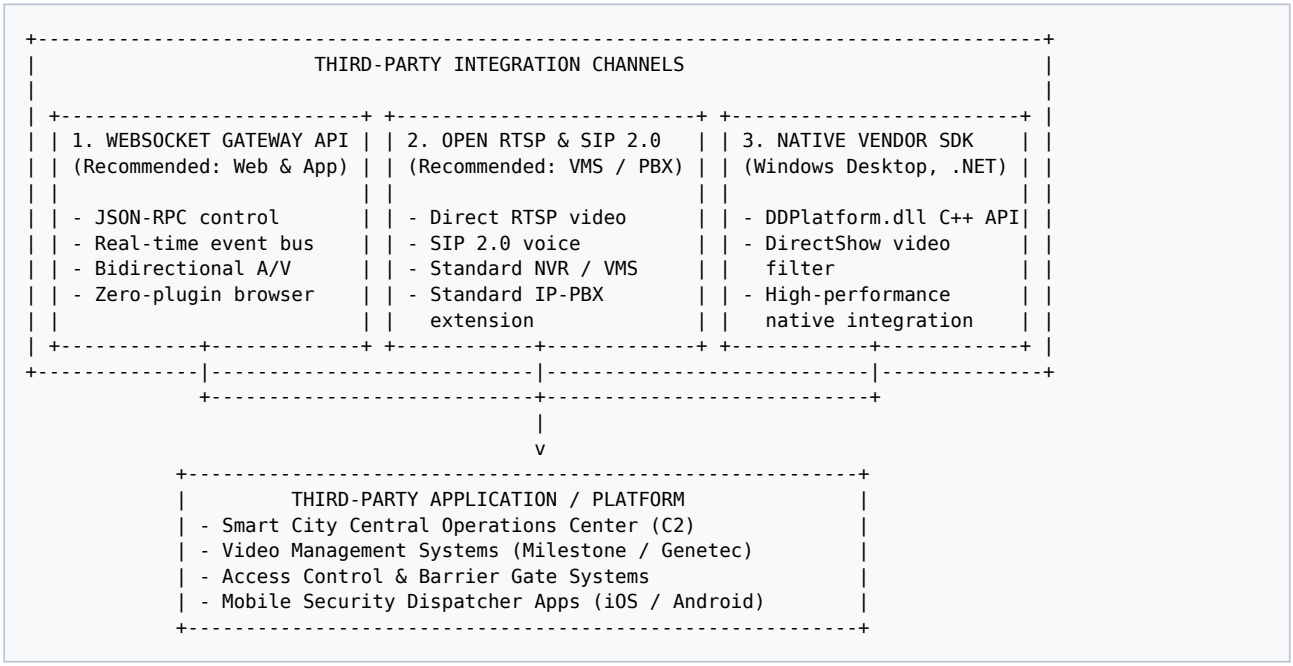


Figure 2. Third-party integration channels

Approach 1: WebSocket Integration Gateway (Recommended for Web and Mobile)

- **Mechanism:** custom applications connect over standard WebSocket (`ws://` or TLS-secured `wss://`) to the gateway service on port `8088` (or `8443` via reverse proxy).
- **Control Channel:** communication uses envelope-wrapped JSON-RPC text frames. Third-party applications send structured commands (e.g., `systemManager.login`, `deviceManager.playRT`, `deviceManager.doControl`) and receive asynchronous push events (e.g., `eventManager.notify` when an SOS button is pressed).
- **Media Channels:** separate binary WebSocket connections deliver raw H.264 video frames and 8 kHz linear PCM audio for both listening (downlink) and talking (uplink).
- **Ideal For:** modern web applications (React, Vue, Angular), mobile applications (Flutter, React Native), and backend microservices (Node.js, Python, Go).

Approach 2: Direct Open Protocols (RTSP and SIP 2.0)

- **Video Surveillance (RTSP):** each intercom camera stream can be ingested directly into third-party Video Management Systems (such as Milestone XProtect, Genetec Security Center, Network Optix, or commercial NVRs) over standard RTSP URLs.
- **Telephony (SIP 2.0):** terminals can be registered as standard SIP extensions with enterprise IP-PBX systems (such as Asterisk, FreeSWITCH, Cisco Unified Communications Manager, or Avaya), allowing security guards to answer emergency calls from standard desk VoIP phones.
- **Ideal For:** existing CCTV / VMS platforms and enterprise telephony systems.

Approach 3: Native Platform SDK (DDPlatform.dll)

- **Mechanism:** direct linking against the vendor's compiled dynamic-link libraries for native Windows desktop environments.
- **Capabilities:** provides low-level control over device telemetry, direct DirectShow video filtering (`DDMediaFilter.dll`), and raw audio buffers.
- **Ideal For:** on-premise Windows desktop dispatch consoles and specialized industrial control software.

7.3 Nature and Ease of Integration Assessment

Integration Domain	Complexity	Implementation Nature and Characteristics
Call Alerts & Event Telemetry	Low (straightforward)	Real-time push events are delivered via standard JSON over WebSockets. Applications listen for `DeviceStatus: "Start"` to trigger emergency popups, audible alarms, or GIS map centering with minimal code.
Relay & Barrier Gate Control	Low (straightforward)	Triggering external barrier gates or door strikes requires sending a single JSON-RPC method (`deviceManager.doControl` with relay index 0 or 1). Execution occurs within milliseconds.
Live Video Ingest	Moderate	Video is delivered as standard H.264 Annex-B frames over WebSocket or RTSP. In web browsers, integration uses standard HTML5 Media Source Extensions (`wfs.js`), requiring no custom browser plugins. In desktop/VMS environments, standard RTSP ingestion is plug-and-play.
Two-Way Voice (Intercom Talk)	Moderate	Voice operates at 8,000 Hz, 16-bit linear PCM (industry standard for telecom audio). Custom web or mobile clients must capture microphone audio, resample to 8 kHz, and packetize into 320-byte (20 ms) frames. Reference blueprints are available to streamline implementation.
Public Address (PA) Broadcast	Low to moderate	Allows one-to-many voice paging across multiple terminals simultaneously without initiating a telephone ring state, using dual-rate 8 kHz / 16 kHz streaming.

7.4 Technical Constraints, Documentation Gaps and Integration Realities

While the system is technically capable of broad integration, enterprise architects and system integrators must account for several operational realities and technical boundaries inherent to the manufacturer's software stack.

1. Absence of Formal Developer Documentation

- The manufacturer does not supply public REST / OpenAPI (Swagger) specifications, developer portals, or modern integration SDK manuals.
- The underlying software was historically conceived as a closed, proprietary appliance designed primarily to pair with the vendor's own legacy desktop client (`AlarmClient.exe`). Integrating with modern third-party web or cloud platforms requires leveraging the gateway translation layer and conforming to undocumented JSON-RPC message structures.

2. Proprietary Media Formatting and Web Demuxing

- The video streams emitted by the gateway are raw H.264 Annex-B byte streams rather than browser-native WebRTC or HLS feeds. Third-party web applications must incorporate a client-side demuxer (such as `wfs.js`) to repackage NAL units into fragmented MP4 (fMP4) chunks for HTML5 `<video>` elements.
- Voice uplink audio requires exact packetization: operator microphone audio must be captured, resampled from system rates (44.1 / 48 kHz) to 8,000 Hz mono 16-bit PCM, and transmitted in strict 320-byte chunks (exactly 20 milliseconds per packet at 50 packets per second). Any deviation causes terminal DSP silence.

3. Proprietary Server Recording Container

- As detailed in Section 5, automatic recordings created by `AlarmServer.exe` are written in a proprietary .bin container format rather than standard MP4. Exporting or streaming these files to non-vendor systems requires format extraction or third-party client-side recording tools.

4. Strict Relational User-Group Permission Binding

- The server enforces an internal multi-tenant permission model. New administrative or dispatcher user accounts created in the database are not permitted to stream video or answer calls by default; they must be explicitly bound to the corresponding “Device Group” node within the server's relational database before media sessions will succeed.

5. Legacy Desktop Operating Architecture

- The core PBX server (`AlarmServer.exe`) is architected as an interactive Windows Win32 application. Deploying the system in enterprise server environments requires managing these processes with system supervision daemons (such as NSSM or Scheduled Tasks) alongside the WebSocket gateway bridge to provide uninterrupted 24/7 background operation.

4.8 Security Hardening and Regulatory Standards

8.1 Security and Network Protection

- **Network Segmentation:** intercom terminals are recommended to be deployed on an isolated, dedicated VLAN, preventing unauthorized cross-traffic from general corporate or public networks.
- **Access Control and Permissions:** the software suite enforces role-based access control (RBAC). Administrators can define specific permissions for each security operator, restricting sensitive actions (such as triggering perimeter barrier relays or accessing historical records) to authorized personnel.
- **Secure Web Access:** modern web console connections support TLS/HTTPS and encrypted WebSockets (WSS), preventing eavesdropping on control or media streams across network links.

8.2 Regulatory and Standards Alignment

The Diihoo IP Emergency Intercom System conforms to relevant international engineering, environmental, and telecommunications standards.

Standard / Specification	Regulatory Alignment and Certification Scope
IEC / EN 60529	Ingress Protection Level IP65 / IP66 (waterproof and dustproof)
IEC / EN 62262	Impact Resistance Level IK08 / IK10 (vandal-proof enclosure)
EN 55032 / EN 55035	Electromagnetic Compatibility (EMC) for commercial equipment
FCC Part 15, Subpart B	Unintentional radiators in commercial and industrial systems
RoHS (2011/65/EU)	Restriction of Hazardous Substances
ISO 9001:2015	Manufacturer quality management system certification
SIP 2.0 / VoIP Architecture	Southbound and central telephony interoperability alignment

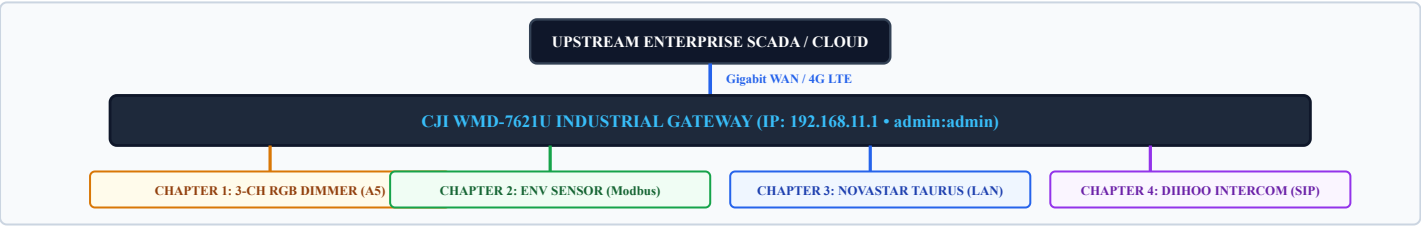
5.1 CJI WMD-7621U Industrial Gateway & Silicon SoC Specifications

The **CJI WMD-7621U Industrial Gateway** (exported by China Jiangsu International / Hi-Wooya IoT platform) serves as the primary communications hub and physical interface bridge across municipal smart-pole, architectural lighting, and environmental monitoring installations. Operating on an embedded MediaTek MIPS silicon core with industrial solid-state storage, the device bridges low-level RS-485 serial field buses directly into high-speed TCP/IP, Ethernet, and cellular wireless cloud networks.

System Subsystem	Component Specification	Operational & Architecture Details
Processor (SoC)	**MediaTek MT7621A Dual-Core MIPS 1004Kc**	Dual-core 32-bit RISC operating @ **800/880 MHz** with 4 Virtual Processing Elements (hardware threads); 32KB I-Cache / 32KB D-Cache per core.
Operating System	**Linux 3.4.113 (Padavan Custom Firmware)**	SMP kernel with low-latency preemption; high-efficiency MediaTek Packet Processing Engine (PPE) hardware NAT offloading; SquashFS root with writable EXT4.
System Memory (RAM)	**128 MB DDR3 SDRAM**	126,144 KB physical memory pool; ~105 MB continuously free for edge daemons and communication buffers under full operational load.
Flash Partitions	**16 MB SPI NOR Flash (8 Partitions)**	`mtd0` : U-Boot (192KB); `mtd1` : NVRAM Config (64KB); `mtd4` : RootFS SquashFS (8.2MB); `mtd5` : Storage (256KB); `mtd7` : **/userSpace (6MB EXT4 writable)**.
Ethernet Switch	**5-Port Gigabit Switch (MT7621 ESW)**	Configured in `WLLLL` topology: **1 Gigabit WAN port (`eth3`) + 4 Gigabit LAN ports (`br0` / `eth2`)** with 1000BASE-T auto-MDIX.
Cellular Modem	**Industrial 4G/5G LTE Subsystem**	Managed by `/usr/bin/atManage` daemon; supports automatic APN provisioning, SIM PIN verification, and real-time AT diagnostic terminals.
Hardware RTC	**Dallas DS1307 Real-Time Clock**	Bit-banged I2C interface routed across GPIO 16 (SCL) and GPIO 17 (SDA); maintains accurate calendar timing across power outages without NTP.
Default Network Access	**IP: 192.168.11.1 • Ports 80, 23, 22**	Default HTTP Web UI on port 80; Telnet root shell on port 23; Dropbear SSH on port 22. **Credentials:** `admin` / `admin` .

5.2 Gateway Operational Role & Subsystem Interconnection Topology

In multi-device deployments, the CJI WMD-7621U functions as the central **protocol translation and Layer-2/3 routing boundary**. Rather than running a monolithic, closed display or PLC application, the gateway establishes high-speed transparent bridging pipelines that interconnect all four field subsystems described in this manual:



5.2.1 RS-485 Half-Duplex Field Bridging

Both the **3-Channel RGB LED Dimmer** and **22-Element Environmental Transmitter** share physical wiring on RS-485 Serial Port 3 (`/dev/ttyS2`). The gateway's `ser2net_s2` daemon transparently converts incoming TCP socket packets on **port 5021** into UART differential pulses on pins A/B, returning raw Modbus and A5 replies without modifying byte payloads.

5.2.2 Gigabit LAN Subsystem Interconnection

The **NovaStar Taurus Multimedia Player** and **Diihoo HD IP Intercom** connect directly into the gateway's onboard LAN ports (`br0` , subnet `192.168.11.0/24`). The gateway provides high-speed hardware-accelerated local packet routing, DHCP leasing via `dnsmasq` , SIP VoIP media transport, and upstream WAN routing to external streaming CDNs or security vaults.

Core Architectural Identity: The CJI Gateway is fundamentally an *Industrial Edge Routing & Transparent Communications Bridge*. It provides transparent Layer-2/Layer-3 connectivity and direct hardware pin manipulation without enforcing rigid, high-level proprietary UI lock-in.

5.3 Gateway Web Management Interface Architecture & Modules

The WMD-7621U runs an optimized, highly responsive Padavan-based embedded HTTP server (`httpd`) accessible at **`http://192.168.11.1/`** . Protected by HTTP Basic Authentication (`admin` / `admin`), the interface provides complete granular control over all router, cellular, serial, and switching functions:

Management Module	Web Path / ASP Script	Configurable Functions & Capabilities
Main Dashboard & Topology	`index.asp` , `state.js`	Real-time CPU and RAM utilization meters, active interface IP bindings, uptime counter, hardware NAT status, and WAN connection status.
LAN & DHCP Server	`Advanced_LAN_Content.asp`	Internal bridge configuration (`br0` : 192.168.11.1/24), `dnsmasq` DHCP IP pool allocation (192.168.11.100–150), static MAC reservations.
WAN & Hardware NAT	`Advanced_WAN_Content.asp`	Dynamic DHCP lease, Static IP, or PPPoE on WAN (`eth3`); MediaTek PPE hardware routing acceleration toggle (LAN/WAN wire-speed offload).
Cellular (4G/5G) Modem	`Advanced_Modem_4G.asp`	Carrier APN credential entry, SIM PIN unlocking, dial-up watchdog, RSSI / RSRP / RSRQ signal monitoring, and **interactive AT command terminal**.
Industrial Serial Server	`ser2net_s1.asp` , `ser2net_s2.asp`	Configuration of UART 1 (Port 2) and UART 2 (Port 3): baud rates (300–115200 bps), framing (8N1), operation mode (TCP Server/Client, UDP).
System Administration & Console	`Advanced_Console_Content.asp`	**Interactive Web Terminal (`apply.cgi`)**: allows direct execution of root Linux commands (e.g. `mtk_gpio` , `ps` , `tcpdump` , `dmesg`).
Startup & Firewall Scripts	`Advanced_Scripts_Content.asp`	Non-volatile startup hook: custom shell commands entered into *"Run After Router Started"* are saved into flash storage (`/etc/storage/started_script.sh`).
Firmware & NVRAM Backup	`Advanced_FirmwareUpgrade_Content.asp`	Full binary firmware flashing (TRX images), factory default NVRAM reset, and automated configuration backup (.tar.gz export).

5.4 Industrial Serial Server Bridging Setup (`ser2net_s2` on Port 5021)

To enable remote acquisition from the environmental sensor and lighting commands to the RGB dimmer over Ethernet/WiFi, the serial port must be bridged using the gateway's native `ser2net_s2` daemon via `ser2net_s2.asp` :

Web Parameter	Required Value	Functional Impact
Open service	YES	Launches `/usr/sbin/ser2net_s2` process on boot.
Transmission mode	**Transparent**	Passes raw bytes without modifying payloads.
Baud rate	`9600` bps	Matches sensor and RGB controller factory clocks.
Data / Parity / Stop	`8 / None / 1`	Standard industrial 8N1 serial UART framing.
Work mode	**TCP Server**	Gateway listens for incoming TCP socket connections.
Listening Port	`5021`	Default external TCP port mapped to `/dev/ttyS2` .
Max TCP Clients	`16`	Allows simultaneous SCADA, local app, and diagnostic sockets.
Custom Heartbeat	DISABLED	**Crucial:** Prevents unsolicited packet injection.

Non-Volatile Flash Persistence Protocol

To ensure the serial bridge survives cold reboots or power outages, parameters are committed to SPI NOR flash (`/dev/mtd5`):

```
nvram set s2_enable=1
nvram set s2_mode=0
nvram set s2_workmode=1
nvram set s2_port=5021
nvram set s2_baudrate=9600
nvram set s2_heart=0
nvram commit
/usr/sbin/s2_manage restart
/sbin/mtd_storage.sh save
```

Executing `mtd_storage.sh save` ensures automatic start within 3s of power application.

Transparent Pass-Through Rule: In Transparent mode (`s2_mode=0`), the gateway acts as an invisible byte conduit. Bytes entering TCP port 5021 are placed onto RS-485 pins A/B within 1.5 ms; sensor reply bytes are instantly buffered and forwarded over the TCP socket.

5.5 Real-Time On-Gateway Hardware Verification Methods

5.5.1 Kernel UART Driver Byte Counters

Executing `cat /proc/tty/driver/serial` reveals physical UART interrupt transactions:

```
0: uart:16550A mmio:0x1E000C00 irq:34 tx:1073 rx:0
2: uart:16550A mmio:0x1E000E00 irq:36 tx:856  rx:3368 RTS|CTS
```

Every poll increments `tx` by 8 bytes; every valid reply increments `rx` by 49 bytes, proving physical copper electrical activity.

5.5.2 Live Packet Sniffing via tcpdump

Sniff port 5021 to view bidirectional hex transactions in real time:

```
tcpdump -i any -nn -X 'port 5021 and (tcp[tcpflags] & tcp-push
Query:  ff03 0001 0016 801a  (8-byte query)
Reply:  ff03 2c02 5c00 0000 00... (49-byte reply)
```

Verifies packet integrity before forwarding to SCADA or external client applications.

5.6 Physical Output Port & GPIO Control Architecture (mtk_gpio)

The CJI WMD-7621U provides switched power outputs designed to energize connected peripherals, streetlight luminaires, and field instruments. The switching logic is driven directly through the MediaTek SoC General Purpose Input/Output (GPIO) controller using `mtk_gpio`:

Physical Port	SoC GPIO Pin	Hardware Driver Circuitry	Switching Logic	Functional State Description
Port 1 (AC Relay)	**GPIO 13**	**Electromagnetic Mechanical Relay**	`1 = ON (Closed)` `0 = OFF (Open)`	Switches mains AC line voltage (220V/110V AC); audible physical contact click; complete galvanic air-gap disconnect.
Port 2	**GPIO 14**	**Solid-State Logic Output (TTL)**	`1 = HIGH (3.3V)` `0 = LOW (0.0V)`	Direct silicon logic output from MT7621 pin; controls external solid-state driver or auxiliary optocoupler circuit.
Port 3	**GPIO 15**	**Solid-State Logic Output (TTL)**	`1 = HIGH (3.3V)` `0 = LOW (0.0V)`	Direct silicon logic output from MT7621 pin; controls external solid-state driver or auxiliary optocoupler circuit.
RJ45 Switch Power	**GPIO 0**	**P-Channel Power MOSFET Gate**	**Inverted Logic:** `0 = ON` • `1 = OFF`	Controls internal DC power rail to Ethernet switch PHY transceivers; power-cycles all 4 LAN RJ45 ports.
Relay Mirror	**GPIO 21**	**Auxiliary Logic Indicator**	`1 = ON` • `0 = OFF`	Synchronized auxiliary signal mirroring Port 1 status for external supervision or front-panel indicator LEDs.

5.7 Hardware Analysis: Why We Can Only Physically Cut Power on Port 1

A primary technical inquiry among field commissioning engineers is **why toggling Port 2 or Port 3 does not disconnect connected AC loads, and why only Port 1 produces an audible physical disconnect**. Comprehensive physical auditing of the gateway PCB and Linux kernel drivers reveals two architectural causes:

5.7.1 Mechanical Relay vs. Solid-State Logic Circuitry

On the physical gateway motherboard, **Port 1 is wired through a heavy electromagnetic mechanical relay**. When GPIO 13 transitions from `1` to `0`, the relay coil de-energizes, physically separating the spring-loaded mechanical contacts with an audible *"click"* and creating a complete galvanic air gap that severs 220V/110V AC mains power.

In contrast, **Ports 2 and 3 do NOT possess onboard mechanical relays**. They are 3.3V/5V logic-level outputs routed directly from the MT7621 silicon pins (GPIO 14 and 15) to solid-state buffer transistors. If AC loads are connected directly across auxiliary terminal block connections labeled 2 or 3, **those terminals are tied straight to internal un-switched auxiliary power rails**. Toggling GPIO 14 or 15 drops the logic pin from 3.3V to 0V, but cannot break high-voltage AC current without an external relay module connected to those pins.

5.7.2 Firmware Boot Initialization & Guardian Watchdog

Because the gateway is designed to host mission-critical field sensors (such as the Environmental Sensor and RGB Controller) on its auxiliary lines, the operating system firmware is hardcoded to ensure those sensors never suffer inadvertent power starvation:

- In `/etc/storage/started_script.sh`, Ports 2 and 3 are explicitly booted to `HIGH` (`P2=1`, `P3=1`) upon startup: `mtk_gpio -w 14 1 && mtk_gpio -w 15 1`.
- The proprietary controller daemon (`rcapp221126`) and its watchdog script (`kaiji.sh`) monitor peripheral power lines to ensure continuous sensor telemetry.
- Port 1 is protected by a continuous background guardian loop in `started_script.sh` that queries `/etc/storage/port1_state` every 5 seconds, restoring its persisted state against external resets.

Engineering Deployment Rule: For direct, onboard high-voltage AC load shedding, **Port 1 (GPIO 13) MUST be utilized**. If load switching is required on Channels 2 or 3, an external industrial DIN-rail relay module must be installed and driven via the GPIO 14/15 logic outputs.

5.8 Strategic Advantages: Why a Decoupled Application Layer is Key

A defining technical characteristic of the CJI WMD-7621U is that **it does not include rigid, hardcoded application screens to visually decode specific third-party sensor registers**. In commercial consumer hardware, this might be viewed as a gap; in industrial engineering and smart infrastructure integration, **this decoupled architecture is the gateway's greatest strategic advantage**:

5.8.1 Universal Interoperability & Zero Vendor Lock-in

Closed commercial appliances (such as the NovaStar Taurus platform analyzed in Chapter 3) embed rigid application layers that enforce vendor-specific software (ViPlex) or paid proprietary clouds (VNNOX). In contrast, the CJI Gateway functions as a **pure, transparent Layer-2/3 industrial pipe**.

Because the gateway does not attempt to parse proprietary register tables inside its firmware, **integrators can connect ANY brand or model of RS-485 device**—weather transmitters, power quality meters, solar inverters, flow valves, or LED dimmers—without requiring firmware rewrites, manufacturer licenses, or cloud subscriptions.

5.8.2 Open Edge Computing & Full System Programmability

The gateway provides complete root access over Telnet/SSH and includes an independent, writable 6MB EXT4 storage partition mounted at `/userSpace`.

Integrators can compile and execute lightweight edge services directly on the MediaTek MIPS processor (or connect via upstream Python/Node services like our `server.py`). This enables custom MQTT publishing, Modbus-to-REST bridging, automated local rules, and zero-latency failover logic to execute autonomously without cloud dependence.

5.9 Architectural Limitations & Field Engineering Workarounds

Observed Limitation	Root Technical Cause	Standard Engineering Workaround / Resolution
No Native Sensor UI Display	Gateway acts as transparent serial bridge (`ser2net`); does not decode arbitrary holding registers.	Deploy custom Python/SCADA polling driver (e.g. `server.py` / `read_full_sensor.py`) to acquire port 5021 and render decoded Celsius, %RH, and dB values.
Only One Internal AC Relay	PCB hardware physically mounts only one electromagnetic relay (GPIO 13); Ports 2 and 3 are 3.3V logic lines.	Use Port 1 for direct AC load breaking. For Port 2/3 high-power switching, wire an external solid-state or DIN-rail relay module triggered by GPIO 14/15.
Single-User Web Admin Lockout	Padavan `httpd` locks `apply.cgi` session state when accessed simultaneously by browser and automated scripts.	Bypass HTTP completely for automated control: utilize lightweight **Telnet socket execution on Port 23** (50 ms execution time, zero session lockout).
Flash State Wipe on Factory Reset	NVRAM configuration partitions (`mtd1`, `mtd5`) are wiped if reset button is held for 10 seconds.	Keep automated deployment scripts (e.g. `persist_ser2net.py`) backed up; re-apply startup hooks in `/etc/storage/started_script.sh`.

5.10 Quick-Reference Field Administration & Diagnostic Checklist

Gateway Connectivity & Login Reference

- **Web Management UI:** `http://192.168.11.1/` (HTTP Port 80)
- **Root Telnet Shell:** `telnet 192.168.11.1 23` (Port 23)
- **SSH Secure Shell:** `ssh admin@192.168.11.1 -p 22` (Port 22)
- **Factory Credentials:** Username: `admin` • Password: `admin`
- **Modbus / RGB Serial Bridge:** TCP Socket `192.168.11.1:5021`

Essential Field Diagnostic Shell Commands

```
# Check RS-485 byte activity on UART 2:
cat /proc/tty/driver/serial | grep '2:'

# Check live state of Port 1 Relay (GPIO 13):
mtk_gpio -r 13

# Sniff Modbus query/response frames live:
tcpdump -i any -nn -X 'port 5021'

# Verify active listening serial daemon:
netstat -tuln | grep 5021
```

Chapter 5 Conclusion: The CJI WMD-7621U Industrial Gateway provides an open, highly resilient, and field-proven hardware foundation for smart-pole infrastructure. By pairing transparent serial bridging on Port 5021 with physical relay control on GPIO 13, it enables seamless enterprise integration without proprietary software boundaries.